

**Regular elements, continued. Application to
algebraic groups**

SGA 3, Exposé XIV

English draft

A. Grothendieck, with an appendix by J.-P. Serre

About this draft

This is an unofficial English translation of Exposé XIV, “Regular elements, continued. Application to algebraic groups” from *Schémas en groupes* (SGA 3). The source is the recomposition edited by P. Gille and P. Polo, PDF `Expo14.pdf` (36 pages), from <https://webusers.imj-prg.fr/~patrick.polo/SGA3/>. The seminar was directed by M. Demazure and A. Grothendieck (IHÉS, 1962–1964).

Statement numbers follow that PDF. A clear typographical slip is corrected in the English and marked with a `% typo:` comment; such slips are listed in the README. A slip that would change the mathematics is left as printed and marked `% typo?:`. Scholarly proofreading remains outstanding.

The translator’s contribution is licensed under the MIT License. The French original remains copyright of its authors and publishers.

Contents

About this draft	iii
Exposé XIV. Regular elements, continued. Application to algebraic groups	1
1. Construction of Cartan subgroups and maximal tori for a smooth algebraic group	1
2. Lie algebras over an arbitrary prescheme: regular sections and Cartan subalgebras	3
3. Subgroups of type (C) of group preschemes over an arbitrary prescheme	9
4. A digression on Borel subgroups	14
5. Relations between Cartan subgroups and Cartan subalgebras	18
6. Applications to the structure of algebraic groups	20
7. APPENDIX: Existence of regular elements over finite fields	23

Exposé XIV. Regular elements, continued. Application to algebraic groups

by A. Grothendieck
with an Appendix by J.-P. Serre

1. Construction of Cartan subgroups and maximal tori for a smooth algebraic group

Theorem 1.1. *Let G be a smooth algebraic group over a field k . Then G admits a maximal torus T , hence a Cartan subgroup $C = \text{Centr}_G(T)$.*

PROOF. By XII 3.2, it amounts to the same thing to find a maximal torus T of G , or a Cartan subgroup C of G . Moreover, since the maximal tori of G are those of G^0 , one may suppose G connected. We distinguish two cases:

1°) The field k is finite. Let $\mathcal{T}$ be the scheme of maximal tori of G (Exp XII 1.10.), which is a smooth scheme over k . Note that G acts on $\mathcal{T}$ by inner automorphisms, and by the conjugacy theorem (XII 6.6 a)), two points of $\mathcal{T}_{\bar{k}}$ rational over $\bar{k}$ are congruent under $G_{\bar{k}}(\bar{k})$. Taking into account that $\mathcal{T}$ is smooth over k , hence $\mathcal{T}_{\bar{k}}$ smooth over $\bar{k}$, it follows that $\mathcal{T}_{\bar{k}}$ is isomorphic to $G_{\bar{k}}/\bar{N}$, where $\bar{N}$ is the stabilizer of an element of $\mathcal{T}_{\bar{k}}(\bar{k})$, i.e. the normalizer of a maximal torus $\bar{T}$ of $G_{\bar{k}}$. Consequently, $\mathcal{T}$ is a “homogeneous space” under the action of the group G . A well-known theorem of Lang (Amer. J. Math. 78, 1956, pp. 555–563) tells us that every homogeneous space under a smooth connected algebraic group over a finite field k admits a rational point. In particular, $\mathcal{T}$ admits a rational point, i.e. G admits a maximal torus T . Q.E.D.

2°) The field k is infinite. We shall use the

Lemma 1.2. *Let G be a smooth algebraic group over a field k . Then G admits a subgroup of type (C) (XIII 6.2).*

By XIII 6.3 it amounts to the same thing to say that $\mathfrak{g}$ contains a Cartan subalgebra $\mathfrak{d}$. This is trivial if k is infinite, for then $\mathfrak{g}$ contains a regular element a , and one will take $\mathfrak{d} = \text{Nil}(a, \mathfrak{g})$. The case of finite k is treated exactly as in proof 1°) above, but requires the preliminary construction of the scheme $\mathfrak{d}$ of Cartan subalgebras of $\mathfrak{g}$ and the fact that the latter is smooth over k , which will be seen below (2.16). To establish 1.1 in case 2°) in which we have placed ourselves, it will in any event suffice to know 1.2 for infinite k . Let us also record:

Lemma 1.3. *Let G be a smooth connected affine algebraic group whose reductive center (XII 4.1 and 4.4) is reduced to the unit group. Then G is nilpotent if (and only if) its Lie algebra $\mathfrak{g}$ is nilpotent.*

This is contained in XII 4.9.

We can now give a method of constructing Cartan subgroups of G (also valid when k is finite, admitting 1.2 in this case). Suppose first G affine. We proceed by induction on

⁰_{xy} version of 1/12/08

$n = \dim G$, the assertion being trivial if $n = 0$. Thus suppose $n > 0$ and the assertion proved for dimensions $n' < n$. Let Z be the reductive center of G , and let

$$u : G \longrightarrow G' = G/Z$$

be the canonical homomorphism. By XII 4.7 c), one has a one-to-one correspondence $C' \mapsto C = u^{-1}(C')$ between Cartan subgroups of G' and Cartan subgroups of G . Thus, replacing G by G' , one may suppose that the reductive center Z of G is reduced to the identity element (for this holds for that of G' by XII 4.7 b)). By 1.3, G admits a subgroup D of type (C). We know that over the algebraic closure $\bar{k}$ of k , $D_{\bar{k}}$ contains a Cartan subgroup of $G_{\bar{k}}$ (XIII 6.6 b)), so every Cartan subgroup of D is a Cartan subgroup of G (XIII, 2.8 a)). Thus one is reduced to finding a Cartan subgroup of D . If $\dim D = \dim G$, i.e. $D = G$, then the Lie algebra of G is a Cartan subalgebra of itself, hence is nilpotent, so by 1.3 G is nilpotent, hence is a Cartan subgroup of itself. If $\dim D < \dim G$, then by the induction hypothesis there exists a Cartan subgroup of D , which is therefore a Cartan subgroup of G , which completes the proof in the case where G is affine. In the general case, let Z be the center of G ; then $G/Z = G'$ is affine (XII 6.1), and for every Cartan subgroup C' of G' , its inverse image C in G is a Cartan subgroup of G (XII 6.6 e)). One is reduced to finding a Cartan subgroup of the smooth affine algebraic group G' , a case which has already been treated. $\square$

Corollary 1.4. *Let G be a smooth group scheme of finite type over an artinian scheme S . Then G admits a maximal torus T , hence a Cartan subgroup $C = \text{Centr}_G(T)$. Every torus S in G is contained in a maximal torus.*

One may indeed suppose S local with residue field k ; then by 1.1, $G_0 = G \times_S \text{Spec}(k)$ admits a maximal torus T_0 ; by IX 3.6 bis and X 2.3, T_0 comes from a torus T of G , which is clearly a maximal torus. The last statement follows by applying the preceding result to the centralizer of S , which is indeed smooth over k (XI 2.4).

Remarks 1.5. a) We shall give below (3.20, 3.21 and XV) variants of 1.4 in the case where S is not assumed artinian.

b) I do not know whether every algebraic group G (not necessarily smooth) over a field k admits a maximal torus. The question only arises in characteristic $p > 0$, and using 1.1 for a smooth quotient group of the form $G' = G/I$, where I is a suitable radicial subgroup of G (for example, the kernel of a sufficiently high power of the Frobenius homomorphism), by taking the inverse image in G of a maximal torus of G' , one is reduced to the case where $(G_{\bar{k}})_{\text{red}}$ is a torus ($\bar{k}$ still denoting the algebraic closure of k). It is easy to see that the answer is affirmative when G is commutative (or more generally nilpotent): then G admits a unique maximal torus, which one may construct, for example, by descent from the maximal torus of $G_{\bar{k}}^*$.

c) In the case where G is affine, and k perfect or G solvable, 1.1 is known and due to Rosenlicht; his proof is very different from that given here.

d) When k is infinite, 1.1 is a consequence of the much more precise result that the scheme $\mathcal{T}$ of maximal tori of G is a rational variety, proved below (6.1). The method is essentially a combination of the proof of 1.1 and the explicit description of the structure of the scheme $\mathfrak{d}$ of Cartan subalgebras of $\mathfrak{g}$. To arrive at the desired result, we must first generalize to the case of an arbitrary base prescheme certain results of XIII 4 to 6 (this is the purpose of the next two sections), and make the preceding construction proving 1.1 more precise, using the fact that every Cartan subgroup of G is contained in one and only one subgroup of type (C) of G (Nos. 4, 5).

* M. Raynaud has given a negative answer to the question raised here; cf. XVII Example 5.9.c).

2. Lie algebras over an arbitrary prescheme: regular sections and Cartan subalgebras

Lemma 2.1. *Let A be a ring, $\mathfrak{d}$ a Lie algebra over A , and for every $s \in \operatorname{Spec}(A)$, let $\mathfrak{d}(s) = \mathfrak{d} \otimes_A k(s)$ be the corresponding Lie algebra over the residue field $k(s)$. One assumes the A -module $\mathfrak{d}$ finitely presented. The following conditions are equivalent:*

- (i) *For every $s \in \operatorname{Spec}(A)$, $\mathfrak{d}(s)$ is nilpotent.*
- (ii) *For every $x \in \mathfrak{d}$, $\operatorname{ad}(x)$ is nilpotent.*
- (iii) *There exists an integer $N \geq 0$ such that for every sequence $x_1, \dots, x_N$ of elements of $\mathfrak{d}$, one has*

$$\operatorname{ad}(x_1) \operatorname{ad}(x_2) \dots \operatorname{ad}(x_N) = 0.$$

When A is a field, the equivalence of (i) and (iii) is the definition of “nilpotent”, and that of (ii) and (iii) is a well-known consequence of Engel’s theorem (Bourbaki, *Groupes et algèbres de Lie*, Chap. I, §4, no. 2). In the general case, one trivially has (iii) $\Rightarrow$ (ii), and (ii) $\Rightarrow$ (i) by the preceding result and the fact that (ii) is stable under passage to a quotient and localization. It remains to prove (i) $\Rightarrow$ (iii). When A is artinian local with maximal ideal $\mathfrak{m}$, let $n > 0$ be an integer such that $\mathfrak{m}^n = 0$, and N an integer such that condition (iii) holds for $\mathfrak{d}(s) = \mathfrak{d} \otimes_A A/\mathfrak{m}$; take $N' = nN$; one sees at once that this integer satisfies (iii). When A is noetherian, there then exist finitely many elements $s_i \in \operatorname{Spec}(A)$ and ideals of definition $\mathfrak{q}_i \subset A_{s_i}$ such that the natural map

$$\mathfrak{d} \longrightarrow \prod_i \mathfrak{d} \otimes_A A_i, \quad \text{where } A_i = A_{s_i}/\mathfrak{q}_i,$$

is injective; then by what precedes there exists for every i an integer N_i satisfying (iii) for the Lie algebra $\mathfrak{d} \otimes_A A_i$, and taking for N the largest of the N_i , one satisfies (iii) for $\mathfrak{d}$. Finally, the general case reduces to the noetherian case by the usual procedure explained in EGA IV 8.

Definition 2.2. Let S be a prescheme, $\mathfrak{d}$ a quasi-coherent Lie algebra over S which is a finitely presented module. One says that $\mathfrak{d}$ is locally nilpotent if for every $s \in S$, the Lie algebra $\mathfrak{d}(s)$ over $k(s)$ is nilpotent. One says that $\mathfrak{d}$ is strictly locally nilpotent if it is locally free, and if on every open subset U of S on which it has constant rank r , its Killing polynomial reduces to $P_{\mathfrak{d}}(t) = t^r$.

Of course, if $\mathfrak{d}$ is a locally free Lie algebra (as a module) over S , one defines its Killing polynomial as a polynomial

$$P_{\mathfrak{d}} \in A[t], \quad A = \Gamma(\underline{\operatorname{Sym}}_{\mathcal{O}_S}(\mathfrak{d})) \simeq \Gamma(W(\mathfrak{d}))$$

where A is the ring of sections of the structure sheaf of the vector bundle $W(\mathfrak{d})$ defined by $\mathfrak{d}$. It is clear that the two notions just introduced in 2.2 are stable under base change and local for the fpqc topology. Note:

Proposition 2.3. *If $\mathfrak{d}$ is strictly locally nilpotent, it is locally nilpotent. The converse is true when $\mathfrak{d}$ is locally free and S reduced.*

The proof is immediate.

Definition 2.4. Let S be a prescheme, $\mathfrak{g}$ a Lie algebra over S which is a locally free module of finite type. Let $\mathfrak{d}$ be a Lie subalgebra of $\mathfrak{g}$; one says that it is a Cartan subalgebra if it satisfies the following conditions:

- (i) $\mathfrak{d}$ is a submodule locally a direct summand (hence also locally free of finite type).
- (ii) For every $s \in S$, $\mathfrak{d}(s)$ is a Cartan subalgebra of $\mathfrak{g}(s)$.

Definition 2.5. Let $S, \mathfrak{g}$ be as in 2.4. A section a of $\mathfrak{g}$ is said to be quasi-regular if for every $s \in S$, $a(s) \in \mathfrak{g}(s)$ is a regular element of the Lie algebra $\mathfrak{g}(s)$ over $k(s)$. One says that a is a regular section if it is quasi-regular and contained in a Cartan subalgebra of $\mathfrak{g}$.

Notions 2.4 and 2.5 are again stable under base change and local for the fpqc topology. Only the last assertion, and for the case of the notion “regular section”, requires a proof, and follows from the fact that the Cartan subalgebra containing a given regular section is uniquely determined. More precisely:

Proposition 2.6. *Let $S, \mathfrak{g}$ be as in 2.4, and let a be a quasi-regular section of $\mathfrak{g}$. Then there exists at most one Cartan subalgebra of $\mathfrak{g}$ containing a . For one to exist, i.e. for a to be a regular section, it is necessary and sufficient that a satisfy the following condition: $\mathfrak{d} = \text{Nil}(a, \mathfrak{g})$ is a submodule locally a direct summand of $\mathfrak{g}$, and $\text{ad}(a)$ induces an automorphism of $\mathfrak{g}/\mathfrak{d}$. In this case, $\mathfrak{d}$ is the unique Cartan subalgebra of $\mathfrak{g}$ containing a .*

Indeed, suppose that a is contained in the Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}$. Then $\text{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ is bijective on every fiber, hence (since $\mathfrak{g}/\mathfrak{d}$ is locally free of finite type) it is an automorphism of $\mathfrak{g}/\mathfrak{d}$; on the other hand, by 2.1, $\text{ad}(a)_{\mathfrak{d}}$ is locally nilpotent, hence $\mathfrak{d} = \text{Nil}(a, \mathfrak{g})$, which proves the uniqueness of $\mathfrak{d}$ and the necessity of the announced regularity criterion. For sufficiency, note that the hypothesis made on a implies that the formation of $\text{Nil}(a, \mathfrak{g})$ commutes with extension of the base and in particular with passage to the fibers, which shows in particular that the fibers $\mathfrak{d}(s)$ of $\mathfrak{d} = \text{Nil}(a, \mathfrak{g})$ are Cartan subalgebras of the $\mathfrak{g}(s)$; moreover $\mathfrak{d}$ is a Lie subalgebra of $\mathfrak{g}$ by XIII 4.1, hence it is a Cartan subalgebra.

Corollary 2.7. *Under the conditions of 2.4, let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$, a a section of $\mathfrak{d}$ which is regular in $\mathfrak{g}$, and u an automorphism of $\mathfrak{g}$. For u to leave $\mathfrak{d}$ invariant, it is necessary and sufficient that $u(a)$ be a section of $\mathfrak{d}$.*

Indeed, then by transport of structure $u(a)$ is a regular section of $\mathfrak{g}$, contained in both Cartan subalgebras $\mathfrak{d}$ and $u(\mathfrak{d})$, which are therefore identical.

Corollary 2.8. *Under the conditions of 2.4, let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$. Then for every $s \in S$ such that $k(s)$ is infinite, there exists an open neighborhood V of s and a regular section a of $\mathfrak{g}$ over V such that $\mathfrak{d} = \text{Nil}(a, \mathfrak{g}|_V)$ (i.e. such that a is a section of $\mathfrak{d}|_V$).*

Indeed, the fact that $k(s)$ is infinite ensures the existence of a regular element a_0 of $\mathfrak{g}(s)$ contained in $\mathfrak{d}(s)$; one may extend it to a section a of $\mathfrak{d}$ over an open neighborhood U of s , and since $\text{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ induces an automorphism of $\mathfrak{g}(s)/\mathfrak{d}(s)$, it follows that, provided one restricts V , $\text{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ is an automorphism, which implies that a is a quasi-regular section of $\mathfrak{g}|_V$ satisfying the desired condition.

Let $\mathfrak{g}$ be as in 2.4; then examination of its Killing polynomial implies at once that the function

$$s \longmapsto \text{nilpotent rank of } \mathfrak{g}(s)$$

on S is upper semicontinuous. We are mainly interested in the case where this function is in fact continuous, i.e. locally constant. Here are some variants of this property:

Proposition 2.9. *Let $S, \mathfrak{g}$ be as in 2.4. Consider the following conditions:*

- (C₀): *The nilpotent rank of the $\mathfrak{g}(s)$ ($s \in S$) is a locally constant function of s .*
- (C₁): *There exists a Cartan subalgebra of $\mathfrak{g}$ locally for the fpqc topology.*
- (C'₁): *As (C₁), replacing “fpqc topology” by “étale topology”.*
- (C₂): *Condition (C₀) holds, and for every S' over S , every quasi-regular section of $\mathfrak{g}_{S'} = \mathfrak{g} \otimes_S S'$ is regular.*
- (C₃): *Every $s \in S$ has an open neighborhood V on which the Killing polynomial of $\mathfrak{g}$ is of the form*

$$t^r(t^{n-r} + c_1 t^{n-r-1} + \cdots + c_{n-r})$$

where for every $s \in V$, $c_{n-r}(s) \in \text{Sym}(\mathfrak{g}(s)^\vee)$ is nonzero.

With this notation, one has the implications

$$(C_3) \Rightarrow (C_2) \Rightarrow (C'_1) \Rightarrow (C_1) \Rightarrow (C_0),$$

and when S is reduced, the five conditions considered are equivalent.

PROOF. Note also that the conditions considered are manifestly stable under base change and local for the fpqc topology.

The implications $(C'_1) \Rightarrow (C_1) \Rightarrow (C_0)$ are trivial, and implication $(C_0) \Rightarrow (C_3)$ when S is reduced is immediate. Note moreover:

Corollary 2.10. *Suppose condition (C_0) holds. Let U be the set of elements of $W(\mathfrak{g})$ which are regular in their fiber. Then U is open; in particular, for every section a of $\mathfrak{g}$ over S , the set V of $s \in S$ such that $a(s) \in \mathfrak{g}(s)$ is regular is open.*

Indeed, the first assertion follows from the second (applied to $\mathfrak{g}_{S'}$ for every base change S' over S). For the latter, since one may here suppose S reduced, hence (C_3) verified, it suffices to examine the Killing polynomial of a in $\mathfrak{g}$.

Implication $(C_2) \Rightarrow (C'_1)$ now follows easily from b) in the following more precise statement:

Corollary 2.11. *Suppose condition (C_2) holds. Then:*

- a) *For every $s \in S$ and every Cartan subalgebra $\mathfrak{d}_0$ of $\mathfrak{g}(s)$ such that $\mathfrak{d}_0$ contains a regular element of $\mathfrak{g}(s)$ (a condition automatically verified if $k(s)$ is infinite), there exists an open neighborhood V of s and a subalgebra $\mathfrak{d}$ of $\mathfrak{g}|_V$ whose fiber at s is $\mathfrak{d}_0$. If S_1 is a subscheme of S containing s , and if one has already extended $\mathfrak{d}_0$ to a Cartan subalgebra $\mathfrak{d}_1$ of $\mathfrak{g} \otimes_S S_1$, then one may find an open neighborhood V of s in S and a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}|_V$ such that $\mathfrak{d} \otimes_V (S_1 \cap V)$ is equal to $\mathfrak{d}_1|_{(S_1 \cap V)}$.*
- b) *For every $s \in S$ such that $\mathfrak{g}(s)$ contains a regular element (a condition automatically satisfied if $k(s)$ is infinite), there exists an open neighborhood V of s and a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}|_V$.*

Statement b) follows from a) by taking $\mathfrak{d}_0 = \text{Nil}(a_0, \mathfrak{g}(s))$, a_0 being a regular element of $\mathfrak{g}(s)$. To prove a), say the second formulation, one considers a regular element a_0 of $\mathfrak{g}(s)$ contained in $\mathfrak{d}_0$, extends it in a neighborhood of s in S_1 to a section of $\mathfrak{d}_1$, and extends the latter to a section of $\mathfrak{g}$ in a neighborhood of s . By 2.10, this section is quasi-regular in an open neighborhood V of s , hence regular by (C_2) , so by 2.6, $\mathfrak{d} = \text{Nil}(a, \mathfrak{g}|_V)$ satisfies the desired conditions.

Remark 2.12. I do not know whether statement 2.11 a) and b) is valid without a hypothesis of existence of regular points (when $k(s)$ is finite).

It remains to prove implication $(C_3) \Rightarrow (C_2)$. Note also the following equivalent form of (C_3) :

(C'_3) One has (C_0) , i.e. the nilpotent rank of the $\mathfrak{g}(s)$ ($s \in S$) is locally constant, and on every open subset V of S where this rank has value r , the Killing polynomial of $\mathfrak{g}|_V$ is divisible by t^r .

One must show that this condition implies that every quasi-regular section a of $\mathfrak{g}$ is regular. Taking 2.6 into account, this is contained in the following lemma (applied to the endomorphism $\text{ad}(a)$ of $\mathfrak{g}$), (iv) $\Rightarrow$ (iii):

Lemma 2.13. *Let A be a ring, M a projective A -module of finite type, and u an endomorphism of M . The following conditions are equivalent:*

- (i) *M is the direct sum of two stable modules M', M'' such that $u|_{M'}$ is nilpotent, and $u|_{M''}$ is an automorphism of M'' .*
- (ii) *There exists an integer $n > 0$ such that $\text{Im } u^n + \text{Ker } u^n = M$.*

- (iii) The nilspace $\text{Nil}(u) = \bigcup_{n>0} \text{Ker } u^n$ is a direct summand of M , and $M = \text{Nil}(u) + u(M)$.

These conditions are implied by the following one (and are equivalent to it when A is reduced):
 (iv) Locally on $\text{Spec}(A)$ (for the Zariski topology) the characteristic polynomial $P_u(t)$ of u can be put in the form

$$t^r(t^{n-r} + c_1 t^{n-r-1} + \cdots + c_{n-r}),$$

where c_{n-r} is invertible.

The equivalence of (i), (ii), (iii) is immediate and is recorded for reference. The fact that (i) implies (iv) when A is reduced follows from the fact that in this case a nilpotent endomorphism of a projective module of rank r has characteristic polynomial t^r , whereas in all cases the characteristic polynomial of an automorphism of a projective module of finite type has as constant term the determinant of u up to sign (locally on $\text{Spec}(A)$), hence an invertible element of A . Finally, to prove (iv) $\Rightarrow$ (i), one notes that M is a module over the polynomial ring $A[t]$, letting t act by u , and the well-known identity

$$P(u) = 0$$

shows that M is annihilated by $PA[t]$, hence can be considered as a module over $A[t]/PA[t]$. Now, writing $P = t^r Q$, where the constant term of Q is invertible, one sees at once that

$$PA[t] = t^r A[t] \cap QA[t],$$

so $A[t]/PA[t]$ decomposes as the product of the rings $A[t]/t^r A[t]$ and $A[t]/QA[t]$, whence a corresponding decomposition of M as a sum of two $A[t]$ -modules, i.e. as a sum of two sub- A -modules M' and M'' stable under u ; this is the decomposition considered in (i).

This completes the proof of 2.13, hence of 2.9. $\square$

Corollary 2.14. *When condition (C_3) of 2.9 holds, the Cartan subalgebras of $\mathfrak{g}$ are strictly nilpotent (2.2).*

The proof is immediate.

Remark 2.15. a) Let us point out that one can prove a converse of 2.14: (C_3) is equivalent to the fact that for every S' over S , every quasi-regular section of $\mathfrak{g}_{S'}$ is regular and every Cartan subalgebra of $\mathfrak{g}_{S'}$ strictly nilpotent, or again every quasi-regular section of $\mathfrak{g}_{S'}$ contained in a strictly nilpotent Cartan subalgebra of $\mathfrak{g}_{S'}$.

b) Unlike the other conditions (C_0) to (C_2) , condition (C_3) is of “infinitesimal nature”; more precisely, when S is locally noetherian, $\mathfrak{g}$ satisfies condition (C_3) if and only if for every base change $S' \rightarrow S$, with S' artinian local (if one wishes, S' the spectrum of an artinian quotient of a local ring of S), $\mathfrak{g}_{S'}$ satisfies the same condition. Likewise, when (C_0) holds, the condition for a section of $\mathfrak{g}$ to be regular is of infinitesimal nature.

c) When $\mathfrak{g}$ is the Lie algebra of a smooth group prescheme of finite presentation over S , then we shall see that conditions (C_0) , (C_1) , (C'_1) , (C_2) on $\mathfrak{g}$ are equivalent (5.2 a)); I do not know what happens in general (except that, even for S artinian local, (C_0) does not imply (C_1)). However, even in the case where $\mathfrak{g}$ comes from a G , and S is artinian local, it is not true in general that (C_2) implies (C_3) , for $\mathfrak{g}$ can be nilpotent without being strictly nilpotent. One obtains an example of this fact by starting from a smooth affine group scheme G over the spectrum S of a discrete valuation ring, such that the Lie algebra of the general fiber is not nilpotent (for example, the general fiber is an adjoint semisimple group), and that of the special fiber is nilpotent (for example, the special fiber being a vector group): then for n sufficiently large, the Lie algebra of $G_n = G \times_S S_n$ is not strictly nilpotent, but it is nilpotent.

Let us still place ourselves under the conditions of 2.4, and let

$$\mathcal{D} : ((\text{Sch}))^\circ_S \rightarrow (\text{Ens})$$

be the functor defined by

$$\mathcal{D}(S') = \text{set of Cartan subalgebras of } \mathfrak{g}_{S'}.$$

Introduce also the functor

$$X(S') = \{(\mathfrak{d}, a) : \mathfrak{d} \text{ is a Cartan subalgebra of } \mathfrak{g}_{S'} \\ \text{and } a \text{ a section of } \mathfrak{d}\}.$$

One thus has two projections $(\mathfrak{d}, a) \mapsto \mathfrak{d}$ and $(\mathfrak{d}, a) \mapsto a$:

$$p : X \longrightarrow \mathfrak{d} \quad \text{and} \quad \psi : X \longrightarrow W(\mathfrak{g}).$$

When (C_0) holds, we also consider the open subset U of regular points of $W(\mathfrak{g})$ (cf. 2.10), and condition (C_2) is then expressed by the fact that the morphism

$$\psi^{-1}(U) \longrightarrow U$$

induced by ψ is an isomorphism (a priori, it is a monomorphism by 2.6). Note that it is trivial that the morphism $p : X \longrightarrow \mathfrak{d}$ is representable by a projection of vector bundles (i.e. for every S -morphism $S' \longrightarrow \mathfrak{d}$, corresponding to a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}_{S'}$, $X \times_{\mathfrak{d}} S'$ is representable by a vector bundle over S' , namely $W(\mathfrak{d})$); hence if $\mathfrak{d}$ is representable, so is X . Now one has:

Theorem 2.16. *Let S be a prescheme, $\mathfrak{g}$ a Lie algebra over S which is a locally free $\mathcal{O}_S$ -module of finite type; suppose condition (C_0) of 2.9 holds.*

- a) The functor $\mathcal{D}$ of Cartan subalgebras of $\mathfrak{g}$ defined above is representable by a quasi-projective prescheme of finite presentation over S . The same holds for the functor X defined above.*
- b) When condition (C_2) of 2.9 holds, $\mathcal{D}$ and X are smooth over S , and the morphism $\psi^{-1}(U) \longrightarrow U$ induced by ψ is an isomorphism.*
- c) Still supposing condition (C_2) holds, let $s \in S$, and let $\mathfrak{d}_0$ be a Cartan subalgebra of $\mathfrak{g}(s)$ corresponding to a point d of $\mathcal{D}$ rational over $k(s)$. Suppose that $\mathfrak{d}_0$ contains a regular point of $\mathfrak{g}(s)$ (a condition automatically verified if $k(s)$ is infinite). Let r be the infinitesimal rank of $\mathfrak{g}(s)$, and n its rank over $k(s)$; then there exists an open neighborhood V of d in $\mathcal{D}$ which is S -isomorphic to an open subset V' of $S[t_1, \dots, t_{n-r}]$.*

PROOF. One may suppose that $\mathfrak{g}$ has constant rank n and constant nilpotent rank r . The assertions made about X in a) and b) follow at once from the assertions made about $\mathcal{D}$ and the fact that X is a vector bundle over $\mathcal{D}$ defined by a locally free module, and are included merely for reference.

a) The functor $\mathcal{D}$ is a subfunctor of the functor $\text{Grass}_{n-r}(\mathfrak{g})$ whose value at S' is the set of locally free quotient modules of rank $n - r$ of $\mathfrak{g}_{S'}$, and it is well known that the latter functor is representable by a projective prescheme smooth over S (cf., for example, Séminaire Cartan 1960/61, Exp. 12, Nos. 2 and 3, whose constructions carry over unchanged to the case of preschemes)^{*(1)}. One is therefore reduced to a relative problem, namely the following: given a locally free quotient module of rank $n - r$ of $\mathfrak{g}$, or, what amounts to the same thing, a locally free submodule $\mathfrak{d}$ of rank r which is locally a direct summand, represent the following functor: $F(S') = \emptyset$ if $\mathfrak{d}$ is not a Cartan subalgebra of $\mathfrak{g}_{S'}$, $F(S') = \{\emptyset\}$ otherwise. In fact, we shall see that F is representable by a subprescheme of finite presentation of S (which will show that $\mathcal{D} \longrightarrow \text{Grass}$ is representable by an immersion of finite presentation, and will finish proving a)). One begins by expressing the condition that $\mathfrak{d}_{S'}$ be a Lie subalgebra of $\mathfrak{g}_{S'}$; one sees immediately that this is expressed by the fact that $S' \longrightarrow S$ factors through a certain closed subprescheme S_1 of S , of finite presentation over S (whose local equations over S are written down immediately using a basis of $\mathfrak{g}$ adapted to the submodule $\mathfrak{d}$). One may therefore suppose that one already has $S = S_1$. One must next express that $\mathfrak{d}_{S'}$ contains, locally for fpqc, a quasi-regular section of $\mathfrak{g}_{S'}$, and for this one considers $V = W(\mathfrak{d}) \cap U$, where U is

^{*}Cf. also EGA I, 2nd edition (to appear in North Holland Publishing Cie).

⁽¹⁾**N.D.E.** See §I.1.3 of M. Demazure and P. Gabriel, *Groupes algébriques*, Masson (1970).

the open subset of regular points of $W(\mathfrak{g})$ (2.10); then the structural morphism $V \rightarrow S$ being smooth and quasi-compact, its image S_1 is an open part of S , and the immersion morphism $S_1 \rightarrow S$ is quasi-compact, i.e. of finite presentation. The condition considered on S' is then expressed by saying that $S' \rightarrow S$ factors through S_1 . Thus one is reduced to the case $S = S_1$, and, using descent theory, to the case where $\mathfrak{d}$ admits a section a which is a quasi-regular section of $\mathfrak{g}$. One must finally express that the section $a_{S'}$ of $\mathfrak{g}_{S'}$ deduced from a satisfies $\text{ad}(a_{S'})_{\mathfrak{g}_{S'}/\mathfrak{d}_{S'}}$ bijective, which again amounts to saying that $S' \rightarrow S$ factors through a certain open subscheme of finite presentation of S , namely S_D , where D is the determinant of $\text{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$. But then one sees immediately that $\mathfrak{d}|_{S_D}$ is a Cartan subalgebra of $\mathfrak{g}|_{S_D}$, so S_D represents the functor F , which proves a).

b) Is immediate by 2.11 a) and XI 1.5. Of course, b) is also a consequence of the more precise statement c).

c) Let a_0 be a regular point of $\mathfrak{g}(s)$ contained in $\mathfrak{d}_0$; extend it to a section a of $\mathfrak{g}$ over an open neighborhood V of s ; one may clearly suppose $V = S$. On the other hand, let M_0 be a complement of the vector space $\mathfrak{d}_0$ in $\mathfrak{g}(s)$; then in an open neighborhood V of S there exists a submodule M of $\mathfrak{g}$, a direct summand of $\mathfrak{g}|_V$, such that $M(s) = M_0$, and one may again suppose $V = S$. Let now $\mathbf{V}$ be the subfunctor of $\mathcal{D}$ such that $\mathbf{V}(S')$ is the set of Cartan subalgebras $\mathfrak{d}'$ of $\mathfrak{g}_{S'}$ satisfying the following two conditions:

1°) $\mathfrak{d}'$ is complementary to $M_{S'}$, and

2°) the unique section of $(a_{S'} + M_{S'}) \cap \mathfrak{d}$ is a regular section of $\mathfrak{g}_{S'}$.

Condition 1°) corresponds to an open subset V_1 of $\mathcal{D}$ (induced by the open subset of $\text{Grass}_{n-r}(\mathfrak{g})$ defined by the same condition 1°); the conjunction of 1° and 2° corresponds to an open subset of V_1 by 2.10 and (C₂). Thus $\mathbf{V}$ is represented by an open subscheme V of $\mathcal{D}$, clearly containing a .

On the other hand, let $\mathbf{V}'$ be the subfunctor of $W(M)$ defined by

$$\mathbf{V}'(S') = \{\text{sections } u' \text{ of } M_{S'} \text{ such that:}$$

(i) $a_{S'} + u'$ is a regular section of $\mathfrak{g}_{S'}$, and

(ii) the unique Cartan subalgebra $\mathfrak{d}'$ of $\mathfrak{g}_{S'}$

containing $a_{S'} + u'$ is complementary to $M_{S'}\}$.

Then condition 1°) corresponds to an open subscheme V'_1 of $W(M)$, namely the inverse image of the open subset U of regular points of $W(\mathfrak{g})$ (cf. 2.10) under the translation morphism $m \mapsto a + m$. The conjunction of (i) and (ii) corresponds to an open subset V' of V'_1 , namely the inverse image of V under the obvious morphism $V'_1 \rightarrow \mathcal{D}$ (associating to u' the unique Cartan subalgebra $\mathfrak{d}'$ of $\mathfrak{g}_{S'}$ containing $a_{S'} + u'$). The restriction of this last morphism to V' is a morphism

$$V' \rightarrow V,$$

which is clearly an isomorphism. This proves c). $\square$

Corollary 2.17. *Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over a field k . Then the scheme $\mathcal{D}$ of Cartan subalgebras of $\mathfrak{g}$ (2.16 a)) is quasi-projective, smooth and irreducible. When $\mathfrak{g}$ contains a regular element (for example when k is infinite), $\mathfrak{d}$ is a rational variety, i.e. its function field is a pure extension of k .*

The fact that $\mathcal{D}$ is irreducible follows from the fact that one has a surjective morphism $\psi^{-1}(U) \rightarrow \mathcal{D}$, and $\psi^{-1}(U)$ is irreducible, being isomorphic to the open subset U of $W(\mathfrak{g})$. The assertion about the function field is an immediate consequence of c).

Remarks 2.18. I do not know whether this conclusion remains true if k is finite, without supposing that $\mathfrak{g}$ contains a regular point; compare 2.12. One can prove that it is so when $\mathfrak{g}$ is the Lie algebra of a smooth algebraic group G over k , at least when $G/\text{radical}$ is an “adjoint” semisimple group, using a result of Chevalley mentioned below (cf. Appendix). It is plausible

that this result remains valid without restriction on G ; for this it would suffice that the cited result of Chevalley be proved for every semisimple algebraic group (not necessarily adjoint).

3. Subgroups of type (C) of group preschemes over an arbitrary prescheme

Theorem 3.1. *Let S be a prescheme, G a smooth S -group prescheme, $\mathfrak{g}$ its Lie algebra (which is a locally free module of finite type over S), and $\mathfrak{h}$ a Lie subalgebra of $\mathfrak{g}$ which is (as a module) locally a direct summand in $\mathfrak{g}$, and such that for every $s \in S$, the geometric fiber $\mathfrak{h}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$. Let a be a quasi-regular section of $\mathfrak{g}$ (2.5). Then*

$$M_a = \underline{\text{Transp}}_G(a, \mathfrak{h})$$

(the subfunctor of G whose points with values in S' are the $g \in G(S')$ such that $\text{ad}(g) \cdot a_{S'} \in \Gamma(S', \mathfrak{h}_S)$) is representable by a closed subprescheme⁽²⁾ of G smooth over S , whose structural morphism to S is surjective.

Consider the canonical morphism

$$\varphi : G \times_S W(\mathfrak{h}) \longrightarrow W(\mathfrak{g})$$

given by $(g, x) \mapsto \text{ad}(g) \cdot x$; then M_a is S -isomorphic to $\varphi^{-1}(a)$, the inverse image of a (considered as a section of $W(\mathfrak{g})$ over S) under φ . It will therefore suffice for the smoothness of M_a that we show that φ is smooth at the points of $G \times_S W(\mathfrak{h})$ over $\text{Im}(a)$; more generally, φ is smooth at every point over a point of $W(\mathfrak{g})$ which is regular in its fiber $W(\mathfrak{g}(s))$ over S . To see this, since the source and target of φ are smooth, hence flat and locally of finite presentation over S , one is reduced to verifying it fiber by fiber, which reduces one to the case where S is the spectrum of an algebraically closed field, G thus being a locally algebraic group over k , $\mathfrak{h}$ a subalgebra of its Lie algebra $\mathfrak{g}$ containing a Cartan subalgebra of $\mathfrak{g}$, and a a regular point of $\mathfrak{g}$. One may clearly suppose (taking into account that φ is a G -morphism) that the point considered in $G \times W(\mathfrak{h})$ is of the form (e, a) . One may clearly suppose G connected, hence of finite type over k , but then our assertion is precisely XIII 5.4. Moreover, the fact that M_a is a closed subprescheme of G (of finite presentation over S) is trivial, since M_a is the inverse image of $W(\mathfrak{h})$ under the morphism $g \mapsto \text{ad}(g) \cdot a$ from G to $W(\mathfrak{g})$. The surjectivity of the structural morphism $M_a \rightarrow S$ also reduces to the case of an algebraically closed base field, but then $\mathfrak{h}$ contains a Cartan subalgebra $\mathfrak{d}$ by hypothesis, which is therefore conjugate to the Cartan subalgebra $\mathfrak{d} = \text{Nil}(a, \mathfrak{g})$ by the conjugacy theorem XIII 6.1 a), hence $\mathfrak{h}$ contains a conjugate of a . This completes the proof of 3.1.

Corollary 3.2. *Let $G, \mathfrak{g}$ be as in 3.1, with G of finite type over S , and let $\mathfrak{k}, \mathfrak{h}$ be two Lie subalgebras of $\mathfrak{g}$, locally direct summands (as modules); suppose that one is under one of the following two hypotheses:*

- a) *For every $s \in S$, the geometric fiber $\mathfrak{k}_{\bar{s}}$ is nilpotent and contains a regular element of $\mathfrak{g}_{\bar{s}}$; the geometric fiber $\mathfrak{h}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$.*
- b) *$\mathfrak{k}$ is a Cartan subalgebra of $\mathfrak{g}$.*

Then $\underline{\text{Transp}}_G(\mathfrak{k}, \mathfrak{h})$ is a closed subprescheme of G smooth over S ; moreover, in case a), its structural morphism to S is surjective.

The fact that the transporter is representable by a closed subprescheme of finite presentation of G is immediate, and is left to the reader. Let us first prove smoothness in case a). Suppose first that there exists a section a of $\mathfrak{k}$ which is quasi-regular in $\mathfrak{g}$. Then it suffices to apply 3.1 and the

⁽²⁾**N.D.E.** One will denote it by $\text{Transp}_G(a, \mathfrak{h})$.

Lemma 3.3. *Under the conditions of 3.2 a), if a is a section of $\mathfrak{k}$ quasi-regular in $\mathfrak{g}$, then one has*

$$\underline{\text{Transp}}_G(\mathfrak{k}, \mathfrak{h}) = \underline{\text{Transp}}_G(a, \mathfrak{h}).$$

Indeed, taking the definitions into account, this amounts to showing that if a is moreover a section of $\mathfrak{h}$, then one has $\mathfrak{k} \subset \mathfrak{h}$. Now, since by hypothesis $\mathfrak{k}$ is locally nilpotent, it follows from 2.1 that $\mathfrak{k} \subset \text{Nil}(a, \mathfrak{g})$; on the other hand, $\text{Nil}(a, \mathfrak{g}) \subset \mathfrak{h}$ for $\text{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$ is injective (being so fiber by fiber by XIII 4.8 b)). Whence the conclusion. In the general case, one reduces to the case where S is affine noetherian by the usual standard procedure, then to the case where S is artinian local (smoothness being a property of infinitesimal nature), and by flat descent to the case where its residue field is infinite, hence the fiber $\mathcal{X}_0$ admits an element regular in $\mathfrak{g}_0$. One lifts this element to an element of $\mathcal{X} = \Gamma(\mathfrak{k})$, which reduces one to the preceding case. Thus one has proved, in case a), smoothness of the transporter; as for the fact that its structural morphism is surjective, it reduces to the case where S is the spectrum of an algebraically closed field, hence where $\mathcal{X}$ contains a regular point of $\mathfrak{g}$, and one applies 3.3 and 3.1.

To prove b), one is reduced by the definition of smoothness (XI 1.1) to proving that if S is affine, S_0 a subscheme defined by a nilpotent quasi-coherent Ideal J , and g_0 an element of $G(S_0)$ transporting $\mathfrak{k}_0$ into $\mathfrak{g}_0$, then g_0 lifts to an element g of $G(S)$ transporting $\mathfrak{k}$ into $\mathfrak{g}$. Now the hypothesis made on g_0 implies that one is under the conditions of a), already treated. This completes the proof.

Of course, when in 3.2 b) $\mathfrak{h}$ satisfies the stronger hypothesis of 3.1, then (and only then) the structural morphism $\underline{\text{Transp}}_G(\mathfrak{d}, \mathfrak{h}) \rightarrow S$ is surjective. Using Hensel's lemma XI 1.10, one concludes from 3.1 and 3.2:

Corollary 3.4. *Under the conditions of 3.1 for G and $\mathfrak{h}$, and supposing G of finite type over S : a) For every quasi-regular section a of $\mathfrak{g}$, there exists, locally for the étale topology, a conjugate of a which is a section of $\mathfrak{h}$. b) For every Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}$, $\mathfrak{d}$ is, locally for the étale topology, conjugate to a subalgebra of $\mathfrak{h}$.*

In particular, when $\mathfrak{h}$ is itself a Cartan subalgebra of $\mathfrak{g}$, one finds:

Corollary 3.5. *Let G be a smooth S -group prescheme of finite type, $\mathfrak{g}$ its Lie algebra, and $\mathfrak{d}, \mathfrak{d}'$ two Cartan subalgebras of G . Then $\underline{\text{Transp}}_G(\mathfrak{d}, \mathfrak{d}')$ is identical to the strict transporter of $\mathfrak{d}$ into $\mathfrak{d}'$, and is a closed subprescheme of G smooth over S , with surjective structural morphism. Locally for the étale topology, $\mathfrak{d}$ and $\mathfrak{d}'$ are conjugate.*

The fact that the transporter is here identical to the strict transporter follows trivially from the fact that $\mathfrak{d}$ and $\mathfrak{d}'$ are locally direct summands in $\mathfrak{g}$ and have the same rank at each point. Thus 3.5 is a particular case of 3.4. In particular, if $\mathfrak{d} = \mathfrak{d}'$:

Corollary 3.6. *Let G be as in 3.5, and let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$. Then $\text{Norm}_G(\mathfrak{d})$ is a closed group subprescheme of G smooth over S , whose Lie algebra is identical to $\mathfrak{d}$.*

Indeed, this last assertion amounts to saying that $\mathfrak{d}$ is its own normalizer in $\mathfrak{d}$, which follows at once from the fact that this is true fiber by fiber.

Corollary 3.7. *Let $G, \mathfrak{g}$ be as in 3.5. Then conditions $(C_2), (C'_1), (C_1)$ of 2.9 are equivalent; in other words, if $\mathfrak{g}$ admits a Cartan subalgebra locally for the fpqc topology, then every quasi-regular section of $\mathfrak{g}$ is regular.*

Indeed, let a be a quasi-regular section; let us prove that it is regular. The question being local for the fpqc topology, one may suppose that $\mathfrak{g}$ admits a Cartan subalgebra $\mathfrak{d}$. By 3.4 a), a is then, locally for the étale topology, conjugate to a section of $\mathfrak{d}$, which reduces one to the case where a is a section of $\mathfrak{d}$, where the conclusion is trivial from the definition.

Definition 3.8. Let G be a smooth group prescheme over a prescheme S . One calls a subgroup of type (C) of G a group subscheme D of G , smooth over S , with connected fibers, such that $\mathfrak{d} = \text{Lie}(D)$ is a Cartan subalgebra (2.4) of $\mathfrak{g} = \text{Lie}(G)$, i.e. such that for every $s \in S$, D_s is a subgroup of type (C) of the algebraic group G_s (XIII 6.2).

Theorem 3.9. *Let G be a smooth S -group prescheme of finite presentation, and $\mathfrak{g}$ its Lie algebra. Then:*

a) *The map*

$$D \longmapsto \mathfrak{d} = \text{Lie}(D)$$

establishes a one-to-one correspondence between subgroups of type (C) of G and Cartan subalgebras of $\mathfrak{g}$.

b) *If D and $\mathfrak{d}$ correspond, one has*

$$\text{Norm}_G(D) = \text{Norm}_G(\mathfrak{d}),$$

which is a closed subscheme of G smooth over S , and one has

$$D = \text{Norm}_G(D)^0 = \text{Norm}_G(\mathfrak{d})^0.$$

c) *Two subgroups of type (C), D and D' , of G are conjugate locally for the étale topology.*

PROOF. Let D be a subgroup of type (C) of G , and $\mathfrak{d}$ its Lie algebra; then $D \subset \text{Norm}_G(\mathfrak{d})$, and by definition 3.8 and 3.6 this is an inclusion of group preschemes smooth over S , inducing an isomorphism on the Lie algebras. Since D has connected fibers, one therefore has $D = \text{Norm}_G(\mathfrak{d})^0$. Thus the map considered in a) is injective; let us prove it is surjective. Let then $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$; then by 3.6, $\text{Norm}_G(\mathfrak{d}) = N$ is a closed group subscheme of G smooth over S , admitting $\mathfrak{d}$ as Lie algebra. Since G is of finite presentation over S , the same holds for N , so (as was noted in XII after 7.3) the union of the connected components of the fibers of N is the underlying set of an open group subscheme N° of N , which is clearly a subgroup of type (C) of G with Lie algebra $\mathfrak{d}$. This proves a); the first assertion b) follows at once, and the formula $D = \text{Norm}_G(\mathfrak{d})^0$ has already been proved. Finally, c) follows from a) and 3.5. $\square$

Corollary 3.10. *Suppose that $\mathfrak{g}$ admits a Cartan subalgebra locally for the fpqc topology (or, what amounts to the same thing by 3.9 a), that G admits a subgroup of type (C) locally for the fpqc topology). Consider the functor $\mathcal{D} : ((\text{Sch}))_S^\circ \longrightarrow (\text{Ens})$ defined by $\mathcal{D}(S') =$ the set of subgroups of type (C) of $G_{S'}$. Then this functor is representable by a quasi-projective prescheme smooth over S , with connected geometric fibers. When S is the spectrum of a field k , hence G a smooth algebraic group over k , and $\mathfrak{g}$ admits a regular point (a condition automatically verified if k is infinite), then $\mathcal{D}$ is a rational variety over k .*

Indeed, by 3.9 a), the functor $\mathcal{D}$ is canonically isomorphic to the functor considered in 2.16; on the other hand, by 3.7, condition (C₂) holds. Thus 3.10 follows from 2.16 and 2.17.

Corollary 3.11. *Let D be a subgroup of type (C) of G , and N its normalizer in G . Then the quotient sheaf G/N is canonically isomorphic to the functor $\mathcal{D}$ of 3.10, hence representable by a quasi-projective prescheme smooth over S , with connected geometric fibers.*

Proposition 3.12. *Let G be an S -group prescheme smooth and of finite presentation over S , H, K two group subschemes smooth and of finite presentation over S , K having connected fibers, and $\mathfrak{g}, \mathfrak{k}, \mathfrak{h}$ the corresponding Lie algebras. One assumes that one of the following two conditions holds for the geometric fibers of the latter:*

a) *For every $s \in S$, $\mathfrak{k}_{\bar{s}}$ contains a regular element of $\mathfrak{g}_{\bar{s}}$, and $\mathfrak{h}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$.*

b) *For every $s \in S$, $\mathfrak{k}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$. Under these conditions, for $H \supset K$ to hold, it is necessary and sufficient that one have $\mathfrak{h} \supset \mathfrak{k}$.*

Of course, one need only prove that if $\mathfrak{h} \supset \mathfrak{k}$, then $H \supset K$. In case b), the inclusion $\mathfrak{h} \supset \mathfrak{k}$ shows that one is in fact under the conditions of a), so it suffices to prove a). Proceeding as in 3.2 a) by reduction to the case of S artinian local, one is reduced to the case where there exists a section a of $\mathfrak{k}$ which is quasi-regular in $\mathfrak{g}$. In this case, proceeding as in XIII 5.5, one is reduced to the following statement:

Corollary 3.13. *Let G be an S -group prescheme smooth and of finite presentation over S , H a group subprescheme of G smooth and of finite presentation over S , $\mathfrak{g}$ and $\mathfrak{h}$ the Lie algebras, and a a section of $\mathfrak{h}$ which is quasi-regular in $\mathfrak{g}$. One assumes that for every $s \in S$, the geometric fiber $\mathfrak{h}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$. Let*

$$M_a = \text{Transp}_G(a, \mathfrak{h}),$$

which is a closed subprescheme of G smooth over S (cf. 3.1), so that M_a^0 (the union of the connected components of the identity element in the fibers of M_a) is an open part of M_a , which we shall equip with the structure induced by M_a . One then has $H^0 = M_a^0$.

Clearly, one has $H \subset M_a$, whence $H^0 \subset M_a^0$. Since this is an inclusion of preschemes smooth over S , to prove that it is an equality, one is reduced to verifying it on the fibers, which reduces one to the case where S is the spectrum of an algebraically closed field, a case seen in XIII 5.4.

Corollary 3.14. *Let G be a smooth S -group prescheme of finite presentation, D a subgroup of type (C) of G , and H a group subprescheme of G smooth over S and of finite presentation over S , and $\mathfrak{d}$ and $\mathfrak{h}$ their Lie algebras. Then one has*

$$\underline{\text{Transp}}_G(D, H) = \underline{\text{Transp}}_G(\mathfrak{d}, \mathfrak{h}),$$

and this functor is representable by a closed subprescheme of G smooth over S .

Indeed, the identity between the two transporters follows from 3.12 b), which allows one to apply 3.2.

Corollary 3.15. *Let G, H be as in 3.14, and suppose that for every $s \in S$, the geometric fiber $\mathfrak{h}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$. Suppose moreover that $\mathfrak{g}$ admits a Cartan subalgebra locally for the fpqc topology. Then locally for the étale topology, H contains a subgroup of type (C) of G .*

By 3.7 and 3.9 a), G admits a subgroup of type (C) locally for the étale topology, so one may suppose that G admits such a subgroup, say D . Then the hypothesis on $\mathfrak{h}$ also means that the structural morphism of the transporter considered in 3.14 is surjective (taking into account the conjugacy theorem XIII 6.1 a)). One then concludes by Hensel's lemma XI 1.10.

Corollary 3.16. *Let G, H, K be as in 3.12 a); suppose moreover that for every $s \in S$, the geometric fiber $\mathfrak{k}_{\bar{s}}$ is nilpotent (i.e. $\mathfrak{k}$ is locally nilpotent). Then one has*

$$\underline{\text{Transp}}_G(K, H) = \underline{\text{Transp}}_G(\mathfrak{k}, \mathfrak{h}),$$

and this functor is representable by a closed subprescheme of G smooth over S , with surjective structural morphism to S . Locally for the étale topology, H contains a subgroup conjugate to K .

The identity of the two transporters is again contained in 3.12 a); the assertion about its structure is then precisely 3.2 a), and the last assertion of 3.16 is then a consequence of Hensel's lemma.

Remarks 3.17. a) In 3.12 and 3.16, one may replace the hypothesis that K is smooth over S by the following weaker hypothesis: the sheaf $e_K^*(\Omega_{K/S}^1) = \omega_K^1$ of relative 1-differentials along the unit section is locally free. In this way, 3.12 contains XIII 5.5.

b) Let $G, \mathfrak{g}, \mathfrak{h}$ be as in 3.1, with G of finite presentation over S . Then $N = \text{Norm}_G(\mathfrak{h})$ is not necessarily smooth over S along the unit section, or, what amounts to the same thing, there does not necessarily exist a group subscheme H of S smooth over S whose Lie algebra is $\mathfrak{h}$, even if S is the spectrum of a field. When such an H exists, so that one then has (taking H with connected fibers) $N = \text{Norm}_G(H)$, I do not know whether N is smooth over S . In this question, one may clearly reduce to the case where S is artinian local.

Finally, to end this section, let us examine the case where G is “semisimple” over S :

Theorem 3.18. *Let S be a prescheme, G an S -group prescheme smooth over S , whose geometric fibers are “adjoint” semisimple groups, i.e. semisimple with reductive center (XII 4.1 and 4.4) reduced to the unit group. Then the subgroups of type (C) of G are identical to its maximal tori, hence also to its Cartan subgroups (XII 3.1).*

Taking the definitions into account, one is reduced to the case where S is the spectrum of an algebraically closed field, and to proving then that for a maximal torus T of G , the Lie algebra $\mathfrak{t}$ of T is a Cartan subalgebra of $\mathfrak{g}$, that is (taking into account the inequality: nilpotent rank of $\mathfrak{g} \geq \text{nilpotent rank of } G = \dim T = \text{rank } \mathfrak{t} = r$) that there exists $x \in \mathfrak{t}$ with $\dim \text{Nil}(x, \mathfrak{g}) = r$. Since $\mathfrak{t}$ is abelian and a fortiori nilpotent, it amounts to the same thing to say that there exists an $a \in \mathfrak{t}$ such that $\text{ad}(a)_{\mathfrak{g}/\mathfrak{t}}$ is injective (XIII 5.7 a)). Now consider the characters α of T which occur in the representation induced on T by the adjoint representation of G . The structure theory of the semisimple group G (BIBLE, 13 th. 1 a) and th. 3, cor. 2), more precisely of the “big cell” of G , a semidirect product of T and subgroups P_α isomorphic to the additive group $\mathbf{G}_a$, invariant under T and corresponding to the “root” characters of G for the torus T , shows that the eigenspace of $\mathfrak{g}$ for the unit character is precisely $\mathfrak{t}$, and the other eigenspaces have dimension 1, the associated characters α being precisely the roots of G for T . By the calculation of the reductive center of G as the intersection of the kernels of the characters of T which occur in the adjoint representation of G (XII 4.8), one sees that G being adjoint is interpreted by the fact that the roots generate the lattice $M = \text{Hom}(T, \mathbf{G})$. Now a well-known lemma of root theory tells us that every root belongs to a system of simple roots, hence to a basis of the group generated by the roots, and consequently to a basis of the dual M of $T^{(3)}$. One concludes:

Corollary 3.19. *If G is an adjoint semisimple algebraic group over an algebraically closed field k , and T a maximal torus of G , then for every root α of G with respect to T , $\alpha : T \rightarrow \mathbf{G}_m$, the corresponding homomorphism $\alpha' : \mathfrak{t} \rightarrow k$ is nonzero.*

This result is essentially equivalent to theorem 3.18, for, for a $t \in \mathfrak{t}$, $\text{ad}(t)$ is semisimple, and its eigenvalues on $\mathfrak{g}/\mathfrak{t}$ are precisely the $\alpha'(t)$; hence $\text{ad}(t)_{\mathfrak{g}/\mathfrak{t}}$ is injective if and only if the $\alpha'(t)$ are $\neq 0$, and there exists a $t \in \mathfrak{t}$ with this property if and only if all the α' are $\neq 0$.

Corollary 3.20. *Let S be a prescheme, G an S -group prescheme, smooth and of finite presentation over S , with geometric fibers connected reductive algebraic groups (i.e. extensions of a semisimple group by a torus). Then for every $s \in S$ there exists an open neighborhood U of s such that $G|_U$ admits a maximal torus.**

Indeed, we shall see in XVI that the hypothesis just made on G implies that G is affine over S and has locally constant reductive rank; hence (XII 4.7 c)) G admits a reductive center Z , $G' = G/Z$ is a smooth affine group over S whose reductive center is reduced to the unit group, and finally the maximal tori of G and of G' are in one-to-one correspondence. Moreover, one sees at once that the geometric fibers of G' are connected semisimple groups, and moreover adjoint by definition (their reductive center being trivial). Thus it suffices to

⁽³⁾**N.D.E.** $\hat{T}$ has been replaced by T .

*This result, as well as 2.11 on which it rests, generalizes immediately to the case where s is replaced by a finite part of S contained in an affine open subset.

confine oneself to the case where G is semisimple and adjoint, and by 3.18 one is reduced to finding an open neighborhood U of s and a subgroup of type (C) of $G|_U$, or, what amounts to the same thing (3.9 a)), a Cartan subalgebra of $\mathfrak{g}|_U$. Now this is possible if $k(s)$ is infinite, for by 3.7, $\mathfrak{g}$ satisfies condition (C₂) of 2.9, hence one may apply 2.11 b). In fact, statement 3.20 remains valid without supposing $k(s)$ infinite. Indeed, by the preceding proof, it suffices to know that for every adjoint semisimple group G over a finite field k , the Lie algebra $\mathfrak{g}$ of G contains a regular element. Now this statement has been proved by Chevalley (using the properties of the Coxeter element of the Weyl group...); cf. the Appendix below by J.-P. Serre.

Remarks 3.21. a) Statement 3.20 remains valid, with essentially the same proof, replacing G by a closed group subscheme H , smooth over S , having everywhere the same rank as G (for example, a “parabolic subgroup” of G), provided $k(s)$ is infinite. I do not know whether here again the hypothesis that $k(s)$ be infinite is superfluous. One can show that it is so at least if H is parabolic, thanks to the construction of the radical U of H and of the quotient H/U , which reduces one to the semisimple case. Unfortunately, the method of regular elements seems powerless here, for one easily constructs examples (for example, with the projective group and its “standard” Borel subgroup) where the Lie algebra $\mathfrak{h}$ of H contains no regular element.

b) The proof of 3.20 in fact shows a more precise result (invoking 3.9 b)) in the case where $k(s)$ is infinite, namely that every maximal torus T_0 of G_s comes from a maximal torus T over an open neighborhood of s . I do not know whether this statement remains valid when $k(s)$ is no longer supposed infinite, the difficulty clearly coming from the fact that the Lie algebra $\mathfrak{t}_0$ of T_0 generally contains no regular element of the Lie algebra $\mathfrak{g}_0$ of the fiber G_s . An affirmative answer to this problem would imply the following statement (which is proved only in the case of an infinite residue field or when A is separated and complete): Let A be a local ring with residue field k , M an “Azumaya algebra” over A , i.e. an algebra such that M is a free module of finite type over A and $M_0 = M \otimes_A k$ a central simple algebra over k , and D_0 a commutative subalgebra of M_0 separable over k , such that $[M_0 : k] = ([D_0 : k])^2$; then there exists a commutative subalgebra D of M which is a module direct summand in M and such that $D \otimes_A k = D_0$ (?). (Note that giving M is equivalent to giving a principal homogeneous bundle under the projective group $\mathrm{PGL}(n)_A$, whence an “inner” twisted form G of $\mathrm{PGL}(n)$, and the maximal tori of G correspond one-to-one to the commutative subalgebras D of M , étale over A , of rank n over A .)

c) Applying 3.20 to the centralizer of a subtorus Q of G (G a reductive S -group prescheme), one deduces that every such Q is contained, locally for the Zariski topology, in a maximal torus of G .

4. A digression on Borel subgroups

Definition 4.1. Let G be a smooth algebraic group over an algebraically closed field. One calls a Borel subgroup of G a smooth solvable connected algebraic subgroup which is maximal for these properties.

When G is affine, one thus recovers the terminology of BIBLE 6 def. 1. Note immediately that if Z is a smooth connected subgroup of G contained in the center (or, more generally, solvable and invariant), then for every Borel subgroup B of G , the image BZ of $B \times Z$ under the morphism $(b, z) \mapsto bz$ from $B \times Z$ to G is a smooth solvable connected subgroup of G containing B , hence identical to B , so B contains Z , hence B is the inverse image of an algebraic subgroup B' of $G' = G/Z$, and it is immediate that B' is a Borel subgroup of G' . Taking $Z = \mathrm{Centr}(G^0)_{\mathrm{red}}^0$, $G' = G/Z$ is affine (XII 6.1), so the Borel subgroups of G' are conjugate, and for such a B' , G'/B' is a projective variety (BIBLE 6 th. 4). Consequently:

Proposition 4.2. *Let G be as in 4.1. Then the Borel subgroups of G are conjugate. If B is a Borel subgroup, then G/B is a projective variety. The maximal tori of B (resp. the Cartan subgroups of B , G being connected) are maximal tori of G (resp. Cartan subgroups of G).*

It remains to prove the last assertion, and one may clearly suppose $G = G^0$. For Cartan subgroups, it follows from the analogous assertion in G' (BIBLE 6 th. 4 cor. 4) and from XII 6.6 e). For maximal tori, it follows from the preceding assertion, since by XII 6.6 c) the maximal tori of a smooth algebraic group are the maximal tori of its Cartan subgroups.

Corollary 4.3. *Suppose G connected. Then every element of G is contained in a Borel subgroup of G .*

One is reduced to the same statement in G' , which is well known (BIBLE 6 th. 5 d)).

Corollary 4.4. *Let B be a Borel subgroup of G , C a Cartan subgroup of B , and N its normalizer in G ; then $N \cap B = C$.*

Indeed, $N \cap B = \text{Norm}_B(C)$, so one is reduced to showing that when G is connected and solvable, a Cartan subgroup C is its own connected normalizer. Now, with the preceding notation, C is the inverse image of a Cartan subgroup C' of G' , so one is reduced to the case where G is affine. Since one knows that the normalizer of a Cartan subgroup is smooth (C being its own connected normalizer; cf., for example, XII 6.6 c)), it suffices to see that C and N have the same points with values in k , which is precisely BIBLE th. 6 d).

Definition 4.5. Let G be a smooth group prescheme of finite presentation over a prescheme S . One calls a Borel subgroup of G any smooth group subprescheme B of finite presentation of G such that for every $s \in S$, the geometric fiber $B_{\bar{s}}$ is a Borel subgroup of $G_{\bar{s}}$.

Thus this is, as one verifies at once, a notion stable under base change and local for the fpqc topology (for if k' is an algebraically closed extension of an algebraically closed field k , then an algebraic subgroup B of the smooth algebraic group G over k is a Borel subgroup of G if and only if $B_{k'}$ is one of $G_{k'}$). It follows from this definition that if G is a smooth algebraic group over an arbitrary field k , and B a Borel subgroup of G , then G/B is a projective variety, every maximal torus T of B is a maximal torus of G , its normalizer in B is identical to its centralizer C , and is a Cartan subgroup of G when G is connected.

Remarks 4.6. Unfortunately, it is no longer true in general (even if G is affine over S and S is the spectrum of the algebra of dual numbers over an algebraically closed field k) that two Borel subgroups of G are conjugate locally for the fpqc topology. As a consequence of this regrettable fact, let us point out that if G is a smooth affine connected algebraic group over an imperfect field k , it is not in general possible to define naturally a homogeneous space D under G , playing the role of a flag variety, i.e. of the variety of Borel subgroups of G (which over the algebraic closure $\bar{k}$ of k would therefore be isomorphic to $G_{\bar{k}}/\bar{B}$, where $\bar{B}$ is a Borel subgroup of $G_{\bar{k}}$). Indeed, when the quotient of $G_{\bar{k}}$ by its radical $\bar{R}$ is an adjoint semisimple group, then the kernel of $G_{\bar{k}} \rightarrow \underline{\text{Aut}}_{\bar{k}}(\bar{D})$ is the radical of $G_{\bar{k}}$; hence if $\bar{D}$ comes from a homogeneous space D under G , the radical $\bar{R}$ comes from a subgroup R of G . Now one easily constructs examples where $G_{\bar{k}}/\bar{R}$ is adjoint but $\bar{R}$ is not “defined over k ”. It is easy to see that under these conditions, the functor $\mathcal{B} : ((\text{Sch}))_S^\circ \rightarrow (\text{Ens})$ such that $\mathcal{B}(S') =$ the set of Borel subgroups of $G_{S'}$ is not representable by a smooth S -prescheme. From the infinitesimal point of view (III §3), the failure of the conjugacy theorem is expressed by the fact that if B is a Borel subgroup of the smooth algebraic group G , the cohomology group $H^1(B, \mathfrak{g}/\mathfrak{b})^{(4)}$ may be nonzero.

On the other hand, we shall see in a later exposé that when G is semisimple, or more generally reductive, such unpleasant phenomena do not occur. It is doubtless these phenomena,

⁽⁴⁾**N.D.E.** Where $\mathfrak{b}$ is the Lie algebra of B .

as well as the absence of good existence theorems, which explain why Borel subgroups play only a relatively unobtrusive role in the study of general group schemes from the scheme-theoretic point of view, whereas they will dominate the theory of semisimple group schemes in later exposés.

Proposition 4.8. ⁽⁵⁾ *Let G be a smooth S -group prescheme of finite presentation with connected fibers, and B a Borel subgroup of G ; then B is identical to its own normalizer, and it is a closed subprescheme of G .*

Indeed, by XII 7.10, one is reduced to proving that over an algebraically closed field k , every element of $G(k)$ which normalizes B is in $B(k)$, which for affine G is a fundamental result of Chevalley (BIBLE 9 th. 1); the general case reduces to it at once by the reduction already used in 4.2.

Remarks 4.8.1. One may generalize definition 4.5 by introducing also the notion of parabolic subgroup of G : one thus calls a group subprescheme P of G , smooth and of finite presentation over S , such that for every $s \in S$, the geometric fiber $P_{\bar{s}}$ is a parabolic subgroup of $G_{\bar{s}}$, i.e. contains a Borel subgroup of $G_{\bar{s}}$. Proposition 4.8 extends (with the same proof of reduction to the “set-theoretic” statement, which is known) to the case of a parabolic subgroup P of G . Note the following consequence of this result (cf. XVI). If P is a parabolic subgroup of G , then G/P is representable by a quasi-projective prescheme of finite presentation over S (N.B. one assumes G has connected fibers). Moreover, G/P is clearly smooth over S , and moreover has connected proper geometric fibers, whence one may easily conclude, using EGA III 5.5.1, that $D = G/P$ is in fact proper, hence projective, over S . Moreover, if its relative dimension is n , it is known that the invertible sheaf $\Omega_{D/S}^n$ is such that its inverse induces ample sheaves on the geometric fibers of D/S , hence (EGA III 4.7.1) $(\Omega_{D/S}^n)^{-1}$ is ample on D relative to S .

One easily sees, by reduction to the affine case and to the case of an algebraically closed base field, that if $u : G \rightarrow G'$ is an epimorphism of smooth algebraic groups, then for every Borel subgroup B of G , $u(B) = B'$ is a Borel subgroup of G' . We are interested in the case where one thus obtains a bijective correspondence between Borel subgroups of G and of G' :

Proposition 4.9. *Let G, G' be two smooth S -group preschemes of finite presentation with connected fibers, and $u : G \rightarrow G'$ a faithfully flat homomorphism of groups, i.e. surjective⁽⁶⁾. One assumes that one is in one of the following two cases (where one has set $N = \text{Ker } u$):*

a) N is central in G .

b) S is the spectrum of a field k , and if $\bar{k}$ denotes an algebraic closure of it, $N_{\bar{k}} = \text{Ker } u_{\bar{k}}$ is contained in the radical of $G_{\bar{k}}$, i.e. in the largest smooth connected solvable invariant subgroup of $G_{\bar{k}}$.

Then the map $B' \mapsto u^{-1}(B')$ induces a bijection from the set of Borel subgroups of G' onto the analogous set for G .

Case b) follows at once from the correspondence between algebraic subgroups of G' and algebraic subgroups of G containing N , and the fact that when k is algebraically closed, the Borel subgroups of G contain the radical of G (which is immediate by the argument preceding 4.2).

Let us prove case a). By XII 7.12, the map $H' \mapsto H = u^{-1}(H')$ establishes a one-to-one correspondence between group subpreschemes H' of G' which are smooth and of finite presentation over S , with connected fibers, and which have at every $s \in S$ the same reductive rank and the same nilpotent rank as G' , and group subpreschemes H of G having the analogous properties. Now the Borel subgroups (of G' , or of G) have the properties in question. It remains to prove that if H', H correspond, then H' is a Borel subgroup of G'

⁽⁵⁾**N.D.E.** There is no number 4.7.

⁽⁶⁾**N.D.E.** Add a reference here?

if and only if H is one in G . By definition, this question reduces to the case where S is the spectrum of an algebraically closed field. Now, since N is central in G , hence in H , it follows that H is solvable if and only if H' is. Finally, taking into account the correspondence between algebraic subgroups of G' and algebraic subgroups of G containing N , one sees at once that H' possesses the maximality property of definition 4.1 if and only if H possesses it, which completes the proof.

Corollary 4.10. *With the notation of 4.7, if B' and B are Borel subgroups of G' and G which correspond, one has*

$$\mathfrak{b} = \text{Lie}(u)^{-1}(\mathfrak{b}')$$

where $\mathfrak{g}, \mathfrak{g}', \mathfrak{b}, \mathfrak{b}'$ are the Lie algebras of G, G', B, B' , and where $\text{Lie}(u) : \mathfrak{g} \rightarrow \mathfrak{g}'$ is the homomorphism deduced from u .

This statement follows trivially from the definitions and the relation $u^{-1}(B') = B$.

We can now prove the main result of the present section:

Theorem 4.11. *Let G be a smooth algebraic group over an algebraically closed field k , and $\mathfrak{g}$ its Lie algebra. Then $\mathfrak{g}$ is equal to the union of the Lie algebras $\mathfrak{b}$ of the Borel subgroups B of G .*

One may clearly suppose G connected. Let R be the radical of G , and let $G' = G/R$. Then 4.9 b) and 4.10 reduce one to proving theorem 4.11 for G' in place of G , i.e. one may suppose G semisimple. Let then Z be the center of G , identical to the reductive center, and let $G' = G/Z$. The same argument (now using 4.9 a)) reduces one to proving the theorem for G' , i.e. one may suppose G adjoint semisimple. Let B be a Borel of G , T a maximal torus of B , hence of G , and $\mathfrak{b}, \mathfrak{t}$ the Lie algebras. By 3.18, T is a subgroup of type (C) of G , i.e. $\mathfrak{t}$ is a Cartan subalgebra of $\mathfrak{g}$, so the union of the conjugates of $\mathfrak{t}$ is dense in $\mathfrak{g}$ (XIII 5.1 (i) $\Rightarrow$ (vii)). A fortiori the union of the conjugates of $\mathfrak{b}$ is dense in $\mathfrak{g}$. Now let X be the closed subscheme of $G/B \times W(\mathfrak{g})$ whose points with values in k are the (g', x) such that $x \in \text{Ad}(g) \cdot \mathfrak{b}$ (cf. XIII 1). Then the morphism $\psi : X \rightarrow W(\mathfrak{g})$ induced by the second projection is proper since G/B is proper over k ; on the other hand one has just seen that it is dominant, so it is surjective, which proves 4.11.

The only result of the present section that we shall use in the rest of this exposé is the following corollary:

Corollary 4.12. *Let k be an infinite field, G a smooth algebraic group over k , T a maximal torus of G , $\mathfrak{g}$ and $\mathfrak{t}$ the Lie algebras, and $u : G \rightarrow \text{GL}(V)$ a linear representation (V a finite-dimensional vector space over k), whence a representation of Lie algebras $u' : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ making V a $\mathfrak{g}$ -module. Then the minimum of the nullity of $u'(x)$ ($x \in \mathfrak{g}$) is attained for an element $x \in \mathfrak{t}$.*

One is reduced at once to the case where k is algebraically closed. One may clearly suppose G connected, and, dividing G by $(\text{Ker } u)_{\text{red}}$, one may suppose G affine. Using 4.11 and the conjugacy theorem for maximal tori of G , one is reduced to the case where G is moreover solvable. Then G is a semidirect product $T \cdot V$, where V is the “unipotent part” of G , which is a smooth connected unipotent group (BIBLE 6 th. 3). Thus $\mathfrak{g}$ is a direct sum (as a vector space) of the Lie subalgebras $\mathfrak{t}$ and $\mathfrak{v} = \text{Lie}(V)$. By the Lie–Kolchin theorem (BIBLE 6 th. 1), $\mathfrak{v}$ admits a composition series by stable subspaces $\mathfrak{v}_i$ such that $\mathfrak{v}_i/\mathfrak{v}_{i+1} = \mathfrak{w}_i$ has dimension 1. Then for each i , one has an induced representation $u_i : G \rightarrow \text{GL}(\mathfrak{w}_i) = \mathbf{G}_m$ and the corresponding homomorphism of Lie algebras $u'_i : \mathfrak{g} \rightarrow k$, so that for every $x \in \mathfrak{g}$, the nullity of $u'(x)$ is equal to the number of i such that $u'_i(x) = 0$. Since V is unipotent, the u_i are trivial on V , hence the u'_i are trivial on $\mathfrak{v}$, which proves that if $x = t + v$ ($t \in \mathfrak{t}$, $v \in \mathfrak{v}$), then $u'_i(x) = u'_i(t)$ for every i , hence the nullity of $u'(x)$ is equal to that of $u'(t)$. Assertion 4.12 follows at once.

5. Relations between Cartan subgroups and Cartan subalgebras

Applying 4.12 to the adjoint representation of G , one finds:

Theorem 5.1. *Let G be a smooth algebraic group over an infinite field, T a maximal torus of G , and $\mathfrak{g} \supset \mathfrak{t}$ the Lie algebras; then $\mathfrak{t}$ contains a regular element of $\mathfrak{g}$.*

Corollary 5.2. *Let G be a smooth S -group prescheme of finite presentation, and $\mathfrak{g}$ its Lie algebra.*

a) *Conditions (C_0) to (C_2) of 2.9 on $\mathfrak{g}$ are equivalent; in particular, if the infinitesimal rank of the fibers of $\mathfrak{g}$ at the points of S is locally constant, then $\mathfrak{g}$ admits a Cartan subalgebra locally for the étale topology, hence (by 3.9 a)) G admits a subgroup of type (C) locally for the étale topology.*

b) *Let H be a group subprescheme of G smooth and of finite presentation over S , with connected fibers having the same reductive rank as G at each $s \in S$ (for example, H is a maximal torus or a Cartan subgroup of G), and D a subgroup of type (C) of G ; then one has $H \subset D$ if and only if one has $\mathfrak{h} \subset \mathfrak{d}$.*

c) *Suppose condition (C_0) holds, i.e. the infinitesimal rank of G is locally constant. Let H be a group subprescheme of G smooth and of finite presentation over S , with nilpotent connected fibers, having at every point the same reductive rank as G (for example, H is a maximal torus or a Cartan subgroup of G). Then H is contained in one and only one subgroup D of type (C) of G .*

d) *Suppose that G admits a Cartan subgroup locally for fpqc; then so does every subgroup D of type (C) of G .*

PROOF. a) Suppose condition (C_0) holds, and let us prove that the same holds for (C_2) , i.e. that every quasi-regular section a of $\mathfrak{g}$ is regular. One reduces as usual to the case of S affine noetherian, then, the question being “infinitesimal” (2.15 b)), to the case where S is artinian local (then (C_0) is moreover trivially satisfied). One may suppose in addition that the residue field k of S is infinite. Note that by 3.7 it suffices to establish that $\mathfrak{g}$ admits a Cartan subalgebra. Let T be a maximal torus of G ; by 5.1 there exists a quasi-regular element t contained in the Lie algebra $\mathfrak{t}$ of T ; let us prove that it is regular, which will finish the proof. Consider the linear representation of T in $\mathfrak{g}$ induced by the adjoint representation of G ; there thus exists a finite set $(u_i)_{i \in I}$ of characters of T such that $\mathfrak{g}$ decomposes as a direct sum of submodules $\mathfrak{g}_i$ stable under T , T acting on $\mathfrak{g}_i$ by u_i (cf. I §4.7.3). Let $u'_i : \mathfrak{t} \rightarrow A$ be the homomorphism of Lie algebras deduced from $u_i : T \rightarrow \mathbf{G}_m$ (N.B. A denotes the ring of S). Consider the homomorphisms u_{i0} and u'_{i0} deduced from the preceding ones by passage to the fibers, i.e. by the base change $A \rightarrow k$. Let I' be the set of $i \in I$ such that $u'_{i0} \neq 0$, and let $I'' = I - I'$. The fact that t is regular is expressed by the condition $u'_{i0}(t_0) \neq 0$ for every $i \in I'$, hence $u'_i(t)$ invertible for $i \in I'$. The Cartan subalgebra of $\mathfrak{g}_0$ defined by t , i.e. the kernel subspace of the semisimple endomorphism $\text{ad}(t_0)$ of $\mathfrak{g}_0$, is $\sum_{i \in I''} (\mathfrak{g}_i)_0$. Consider

$$\mathfrak{d} = \sum_{i \in I''} \mathfrak{g}_i,$$

then $\text{ad}(t)$ is nilpotent on $\mathfrak{d}$; on the other hand, it is an automorphism of $\mathfrak{g}/\mathfrak{d} \simeq \sum_{i \in I'} \mathfrak{g}_i$. By 2.6, t is therefore regular.

b) Follows from 3.12 a), H satisfying the hypothesis that every geometric fiber of $\mathfrak{h}$ contains a regular element of that of $\mathfrak{g}$, by 5.1.

c) By b), one is reduced to proving that $\mathfrak{h}$ is contained in one and only one Cartan subalgebra of $\mathfrak{d}$. By a), one knows moreover that $\mathfrak{g}$ satisfies (C_2) . One is therefore reduced to the

Lemma 5.3. *Let $\mathfrak{g}$ be a Lie algebra over S which is a locally free module of finite type and satisfies condition (C_2) of 2.9. Let $\mathfrak{h}$ be a Lie subalgebra of $\mathfrak{g}$ satisfying the following conditions: it is a module locally a direct summand, it is locally nilpotent, and for every $s \in S$*

the geometric fiber $\mathfrak{h}_{\bar{s}}$ contains a regular element of $\mathfrak{g}_{\bar{s}}$. Then $\mathfrak{h}$ is contained in one and only one Cartan subalgebra of $\mathfrak{g}$.

(N.B. in the case of interest to us, $\mathfrak{h}$ satisfies the stated conditions: it is locally nilpotent for H has nilpotent fibers, and the condition on regular elements follows from 5.1.) Since 5.3 is local for the fpqc topology, it suffices to prove that at a point $s \in S$ such that $\kappa(s)$ is infinite, there exists an open neighborhood U of s such that existence and uniqueness hold for every base change $S' \rightarrow S$ factoring through U . Take a regular element of $\mathfrak{g} \otimes \kappa(s)$ contained in $\mathfrak{h} \otimes \kappa(s)$; extend it to a section a of $\mathfrak{h}$ over an open neighborhood U of s ; by (C₂) one may suppose that this section is regular (2.10). One may suppose $U = S$. A Cartan subalgebra of $\mathfrak{g}$ containing $\mathfrak{h}$ contains a , hence is identical to $\mathfrak{d} = \text{Nil}(a, \mathfrak{g})$ (2.6), whence uniqueness. Moreover, since $\mathfrak{h}$ is locally nilpotent, one indeed has $\mathfrak{h} \subset \mathfrak{d}$, which proves existence.

d) This is a particular case of XII 7.9 d). $\square$

Corollary 5.4. *Let G be a smooth connected algebraic group over an algebraically closed field k , H a connected algebraic subgroup such that $\mathfrak{h}$ contains a Cartan subalgebra of $\mathfrak{g}$, i.e. such that H contains a subgroup of type (C) of G . Then the number of conjugates of H containing a regular element of $G(k)$ is equal to the number of conjugates of $\mathfrak{h}$ containing a regular element of $\mathfrak{g}$.*

Indeed, let g be a regular element of $G(k)$, and C the unique Cartan subgroup of G containing g (XIII 2); then the conjugates of H containing g are those containing C (XIII 2.8 b)). Likewise, let x be a regular element of $\mathfrak{g}$; then if a conjugate $\mathfrak{h}'$ of $\mathfrak{h}$ contains x , then $\text{ad}(x)_{\mathfrak{g}/\mathfrak{h}'}$ is injective (XIII 5.4), so $\mathfrak{h}'$ contains $\text{Nil}(x, \mathfrak{g}) = \mathfrak{d}$; hence the number of conjugates of $\mathfrak{h}$ containing x is equal to the number of conjugates containing the Cartan subalgebra $\mathfrak{d}$. Moreover, $\mathfrak{d}$ is the Lie algebra of a subgroup D of type (C) of G . One may clearly suppose $C \subset D$, and assertion 5.4 will follow from this: for H to contain C , it is necessary and sufficient that $\mathfrak{h}$ contain $\mathfrak{d}$. Indeed, by XIII 5.5 the relation $\mathfrak{h} \supset \mathfrak{d}$ implies $H \supset D$, and a fortiori $H \supset C$. Conversely, by 5.1, $\mathfrak{t}$ contains a regular element x of $\mathfrak{g}$; then $H \supset C$ implies $\mathfrak{h} \ni x$, hence, as already noted, this implies $\mathfrak{h} \supset \mathfrak{d}$. This completes the proof.

Let G be as in 5.2, and suppose that the infinitesimal rank of the fibers of G remains locally constant (condition (C₀)). Then by 5.2 c), one finds a homomorphism of functors on $((\text{Sch}))_S^\circ$:

$$\mathcal{C} \longrightarrow \mathcal{D}$$

where

$$\mathcal{C}(S') = \text{set of Cartan subgroups of } G_{S'},$$

$$\mathcal{D}(S') = \text{set of subgroups of type (C) of } G_{S'}.$$

By 3.10 and 5.2 a), $\mathcal{D}$ is representable by a smooth quasi-projective prescheme over S . Consider the subgroup D of type (C) of $G_{\mathcal{D}}$ playing the universal role relative to G/S ; one may then consider the functor $\mathcal{C}_D : ((\text{Sch}))_{\mathcal{D}}^\circ \rightarrow (\text{Ens})$ defined in terms of the $\mathcal{D}$ -group D as $\mathcal{C}$ in terms of the S -group G . One then has the following result:

Proposition 5.5. *Under the preceding conditions, consider $\mathcal{C}$ as a functor over the prescheme $\mathcal{D}$; then $\mathcal{C}$ is $\mathcal{D}$ -isomorphic to the functor $\mathcal{C}_D$ “of Cartan subgroups of D ”.*

This follows at once from the

Corollary 5.6. *Let G be a smooth S -group prescheme of finite presentation, and D a subgroup of type (C) of G . Then there is a one-to-one correspondence between the Cartan subgroups of G contained in D and the Cartan subgroups of D (more precisely, for a subgroup H of G , H is a Cartan subgroup of G if and only if it is a Cartan subgroup of D).*

Indeed, this is a particular case of XII 7.9 c), taking into account that over an algebraically closed field a subgroup of type (C) of G contains a Cartan subgroup of G .

For the next section, the main result obtained here is 5.2 c) for H a Cartan subgroup of G , which allows one to state 5.5 and thus provides a useful “dévissage” of $\mathcal{C}$.

6. Applications to the structure of algebraic groups

Theorem 6.1. *Let G be a smooth algebraic group over a field k . Consider the scheme $\mathcal{T}$ of maximal tori of G , isomorphic to the scheme $\mathcal{C}$ of Cartan subgroups of G , which is a homogeneous space under G^0 , and a smooth connected affine algebraic scheme (XII 7.1 d)). Then $\mathcal{C}$ is a rational variety, i.e. the field of rational functions on $\mathcal{C}$ is a pure extension of k .*

PROOF. We shall first give the proof in the case where k is infinite. One may clearly suppose G connected, for $\mathcal{T}$, hence $\mathcal{C}$, does not change when G is replaced by G^0 . Moreover, by XII 7.6, $\mathcal{C}$ does not change when G is divided by a central subgroup. This allows us, first dividing by the center of G , to suppose G affine (XII 6.1), then, dividing by its reductive center (XII 4.1 and 4.4), to suppose that the reductive center of G is trivial (XII 4.7 b)). Furthermore, we proceed by induction on $n = \dim G$, supposing the theorem proved for dimensions $n' < n$. If G is nilpotent, then $\mathcal{C}$ is reduced to a point rational over k , and 6.1 is trivial. Otherwise, the Lie algebra of G is not nilpotent (1.3), hence the Cartan subalgebras of $\mathfrak{g}$ have dimension $n' < n$, hence the subgroups of type (C) of G have dimension $n' < n$. Consider then the morphism

$$\mathcal{C} \longrightarrow \mathcal{D}$$

considered in 5.5. One knows by 3.10 (k being an infinite field, hence $\mathfrak{g}$ containing a regular element) that $\mathcal{D}$ is a rational variety, i.e. the field K of rational functions on $\mathcal{D}$ is a pure extension of k . Consider the fiber of $\mathcal{C}$ at the generic point x of $\mathcal{D}$; by 5.5 it is the scheme of Cartan subgroups of a certain smooth connected algebraic group D_x over $K = \kappa(x)$ (namely $D_x =$ “the generic subgroup of type (C) of G ”). The field L of rational functions on $\mathcal{C}$ is therefore isomorphic to the field of rational functions on $\mathcal{C}_{D_x}$, which by the induction hypothesis (since $\dim D_x = n' < n$) is a pure extension of K . Thus by transitivity L is a pure extension of k .

When k is finite, a different proof is needed. One may again suppose that G is affine and connected. Note that k is perfect; it follows at once that the radical $\bar{R}$ of $G_{\bar{k}}$ is “defined over k ”, i.e. comes from a subgroup R of G . Suppose first that $R \neq G$, i.e. G is not solvable, and let

$$u : G \longrightarrow G' = G/R$$

be the canonical morphism. Consider the corresponding morphism $C \longmapsto u(C)$

$$v : \mathcal{C}_G \longrightarrow \mathcal{C}_{G'}$$

(whose definition is immediate by XII 7.1 e)). Let x be the generic point of $\mathcal{C}_{G'}$; then the fiber $v^{-1}(x)$ is identified with the scheme of Cartan subgroups of G_K (where $K = \kappa(x)$) whose image in G'_K is a certain Cartan subgroup C'_x (namely, “the generic Cartan subgroup of G' ”). It is therefore also the scheme of Cartan subgroups of $H = u_K^{-1}(C')$ (XII 7.9 c)), and since K is here an infinite extension of k , it follows from the part of 7.1 already proved that the field L of rational functions on $\mathcal{C}_G$, equal to that of $\mathcal{C}_H$, is a purely transcendental extension of K . To prove that it is a purely transcendental extension of k , it therefore suffices to prove that this is so for K , i.e. one is reduced to the case where G is semisimple. One may moreover suppose that G is adjoint (dividing G by its reductive center). But then by 3.18 one has $\mathcal{C}_G \simeq \mathcal{D}_G$, and by 3.10 it suffices to prove that $\mathfrak{g}$ admits a regular point, which (as was noted in 3.20) is an unpublished result of Chevalley⁽⁷⁾.

It therefore only remains to treat the case where k is finite and G connected affine solvable. One in fact has a more general result:

⁽⁷⁾**N.D.E.** Cf. the Appendix, by J.-P. Serre.

Corollary 6.2. *Let G be a smooth solvable algebraic group over a field k ; then the variety $\mathcal{C}$ of Cartan subgroups of G is isomorphic to an affine space $\text{Spec } k[t_1, \dots, t_N]$.*

One may again suppose G connected and affine. Let G_∞ be the smallest of the groups occurring in the descending central series of G (by G_i such that $G_{i+1} = [G, G_i]$); it is thus the smallest invariant algebraic subgroup of G such that G/G_∞ is nilpotent. Let C be a Cartan subgroup of G (one exists by 1.1); then the image of C in G/G_∞ is a Cartan subgroup of it, and is therefore equal to G/G_∞ ; consequently the morphism $g \mapsto \text{Ad}(g) \cdot C$ from G_∞ to $\mathcal{C}$ is an epimorphism, and identifies $\mathcal{C}$ with the homogeneous space $G_\infty/N \cap G_\infty$, where N is the normalizer of C in G (moreover equal to C , as recalled in 4.4, but that does not matter here). Note that $U = G_\infty$ is clearly a smooth connected “unipotent” algebraic group (for over the algebraic closure of k , it is contained in the unipotent part of G , by the known structure of smooth solvable affine groups, BIBLE 6 th. 3). When k is perfect (the only case needed to establish 6.1), it follows very easily that U is even k -unipotent, i.e. admits a composition series by algebraic subgroups U_i such that U_i/U_{i+1} is isomorphic to $\mathbf{G}_a$. In fact, Rosenlicht has proved that this result remains valid for a group of the form G_∞ as above, without restriction on k (M. Rosenlicht, *Questions of rationality for solvable algebraic groups over non perfect fields*, Annali di Matematica 1963, pp. 97–120, theorem 4 cor. 2), a considerably more delicate result which we shall admit here. It now suffices to apply the following lemma, doubtless well known to specialists:

Lemma 6.3. *Let U be a smooth connected algebraic group over a field k , $X = U/V$ a homogeneous space under U having a point rational over k . Suppose U k -unipotent. Then as a k -scheme, X is isomorphic to an affine space $\text{Spec } k[t_1, \dots, t_N]$.*

Indeed, let $(U_i)_{0 \leq i \leq n}$ be a composition series of U by smooth connected subgroups, with $U_n = (e)$, $U_0 = U$, $U_i/U_{i+1} \simeq \mathbf{G}_a$, the U_i invariant in U . Then the $K_i = U_i V$ are algebraic subgroups of U (not necessarily smooth or connected if V is not, but that matters little to us), and K_{i+1} is invariant in K_i . One has a canonical morphism $U_i/U_{i+1} \rightarrow K_i/K_{i+1}$ which is an epimorphism, which proves that K_i/K_{i+1} is either reduced to the unit group or isomorphic to $\mathbf{G}_a/H_i$, where H_i is a finite subgroup of $\mathbf{G}_a$, which by Rosenlicht (*loc. cit.* th. 2) implies that $K_i/K_{i+1} \simeq \mathbf{G}_a$ (a result moreover immediate if k is perfect). Set now $X_i = U/K_i$; let us prove by induction on i that X_i is isomorphic to an affine space. Indeed, if this holds for X_i , let us prove that the same holds for X_{i+1} . If $K_i/K_{i+1} = e$, one has $X_i = X_{i+1}$, and this is trivial. Otherwise, X_{i+1} is a principal bundle with base X_i and structural group $\mathbf{G}_a \simeq K_i/K_{i+1}$. Thus X_i being affine, X_{i+1} is a trivial bundle, hence isomorphic to $X_i \times \mathbf{G}_a$, which again proves that X_{i+1} is isomorphic to a standard affine space. This proves 6.2 and consequently finishes proving 6.1. $\square$

Corollary 6.4. *Let G be a smooth algebraic group over an infinite field k . Then the set of points of $\mathcal{C}$ (notation of 6.1) rational over k is dense for the Zariski topology. The union of the Cartan subgroups of G is dense in G .*

The first assertion holds for every unirational variety over an infinite field, and is moreover here for us the most important “arithmetic” consequence of the unirationality results. The second assertion follows from the first and from the density theorem XIII 2.1.

Corollary 6.5. *Let G be a smooth connected algebraic group over k . Then the variety Z of regular semisimple points of G (XIII 3.5) is a unirational variety. In particular, if k is infinite, the set of points of Z rational over k is dense in Z .*

Indeed, Z is an open subset of a torus over $\mathcal{C}$, hence its function field L is the function field of a torus defined over the function field K of $\mathcal{C}$ (namely the “generic maximal torus” of G); it is therefore a unirational extension of K (XIII 3.4), and since K is a pure extension of k by 6.1, L is a unirational extension of k .

Corollary 6.6. *Let G be a smooth connected algebraic group over k , and let H be the subgroup of G generated by the subvariety Z of regular semisimple points, i.e. (XII 8.2) the smallest invariant algebraic subgroup of G such that G/H has reductive rank zero (i.e. over the algebraic closure of k is an extension of an abelian variety by a smooth connected unipotent group). Then H is a unirational variety. In particular, if $G = H$, i.e. (XII 8.4) if G is affine and if over the algebraic closure $\bar{k}$ there exists no nontrivial homomorphism from $G_{\bar{k}}$ to $\mathbf{G}_a$, then G is a unirational variety, hence if k is infinite, the set of its points rational over k is dense.*

This follows at once from 6.5, for it is immediate that if one has smooth connected k -preschemes Z_i which are unirational varieties and morphisms $u_i : Z_i \rightarrow G$, then the algebraic subgroup of G generated by the u_i (VI_B 7.1) is a unirational variety.

As an interesting particular case of 6.5 or 6.6 (as one prefers), note:

Corollary 6.7. *Let G be a smooth connected affine algebraic group of unipotent rank zero; then G is a unirational variety.*

One may make 6.4 more precise as follows:

Corollary 6.8. *Let G be a smooth connected algebraic group over the infinite field k , and C a Cartan subgroup of G . Then the union of the conjugates of C by regular semisimple elements of $G(k)$ is dense in G .*

This follows at once from 6.4 and XIII 3.6, which says that the morphism $\varphi : Z \times C \rightarrow G$ defined by $\varphi(t, c) = \text{ad}(t)c$ is dominant. This result also implies (without supposing k infinite):

Corollary 6.9. *Let G be a smooth connected algebraic group over k , H a smooth connected algebraic subgroup of G such that H has the same reductive rank and the same nilpotent rank as G (i.e. over the algebraic closure $\bar{k}$ of k , $H_{\bar{k}}$ contains a Cartan subgroup of $G_{\bar{k}}$). If H is a unirational variety, so is G . If $H(k)$ is dense in H , $G(k)$ is dense in G .*

Indeed, the morphism $\varphi : Z \times H \rightarrow G$ defined by $\varphi(t, h) = \text{ad}(t)h$ is dominant; now by 6.5, Z is a unirational variety, and by hypothesis so is H , hence $Z \times H$, whence the first result. The second is proved similarly.

We now recover the following well-known result (due to Chevalley in characteristic 0, to Rosenlicht in characteristic $p > 0$):

Corollary 6.10. *Let G be a smooth connected affine algebraic group over a perfect field k . Then G is a unirational variety, hence if k is moreover infinite, $G(k)$ is dense in G .*

Indeed, by 1.1, G admits a Cartan subgroup C . By 6.9 it suffices to prove that the latter is a unirational variety. Now k being perfect, one sees at once by Galois descent from the case of algebraically closed k (BIBLE 6 th. 2) that one has $C = T \times C_u$, where T is the maximal torus of C and C_u a smooth connected unipotent group. One already knows that T is a unirational variety (XIII 3.4); it remains to see that the same holds for C_u . Now k being perfect, C_u is even k -unipotent, and one may apply 6.3.

Remarks 6.11. a) There are known examples (Rosenlicht) of twisted forms of $\mathbf{G}_a$ over an imperfect field which have only finitely many rational points, hence a fortiori are not unirational varieties. On the other hand, Chevalley has given an example of a torus over a field of characteristic zero which is not a rational variety. On the other hand, it follows from Chevalley's theory of semisimple groups that over an algebraically closed field, every smooth connected affine algebraic group is a rational variety. Note moreover that the question of unirationality arises in any event only for affine algebraic groups, a unirational algebraic group being necessarily affine by Chevalley's structure theorem.

b) With the notation of 6.6, it is tempting to try to give a condition of unirationality of G in terms of the group G/H (which is unipotent if G is assumed affine). The latter must clearly

be unirational; is this condition also sufficient? Note that an example of Rosenlicht (*loc. cit.*) shows that a smooth connected unipotent algebraic group U can be a rational variety without being k -unipotent.

c) It would be interesting to study, over a finite field k , questions of the “density” type, such as the following (raised by Rosenlicht): Let G be a smooth connected algebraic group over k ; is G then generated by its Cartan subgroups^{*(8)?}

In this question, one can reduce to the case of affine G , by dividing by the center. The answer would be affirmative in the semisimple case if one could make the result of Chevalley on existence of regular points mentioned in 3.20 more precise, so as to obtain a regular element of $\mathfrak{g}$ which does not belong to $\mathfrak{h} = \text{Lie}(H)$, where H is a smooth algebraic subgroup of G with $H \neq G$ given in advance (G an adjoint semisimple group over the finite field k).

7. APPENDIX: Existence of regular elements over finite fields

by J.-P. Serre

Throughout what follows, k denotes a finite field and $\bar{k}$ its algebraic closure; the Galois group of $\bar{k}/k$ is denoted by $\mathcal{G}$; recall that if $q = \text{Card}(k)$, $\mathcal{G}$ is topologically generated by the “Frobenius” element $F : x \mapsto x^q$.

One proposes to prove the following theorem⁽¹⁾:

Theorem. *Let G be an adjoint semisimple group defined over k , and $\mathfrak{g}$ its Lie algebra. The k -Lie algebra $\mathfrak{g}(k)$ contains a regular element.*

Remarks. (1) It is well to recall that “adjoint” means that the center of G is trivial (as a group subscheme of G). In view of the Chevalley seminar, this also means that if T is a maximal torus of G , the group $X(T)$ of characters of T (defined over $\bar{k}$) is generated by the set R of roots. It follows in particular that the rank of the Lie algebra $\mathfrak{g}$ is equal to the dimension of T (i.e. to the rank of G).

(2) The writer does not know whether the hypothesis that G is adjoint is indispensable.

Lemma 1. *It suffices to prove the theorem when G is geometrically simple.*

^{*}This question has since been answered in the affirmative by Steinberg.

(8)**N.D.E.** Having been unable to identify this result in the *Collected Papers* of R. Steinberg, let us give a proof based on the Bruhat decomposition. Let G/k be a semisimple group defined over the finite field k . One must show that G is generated by its k -Cartan subgroups. This question is stable under central isogeny and restriction of scalars; one is therefore reduced to the case where G is geometrically simple (in the same way as in lemma 1 of the appendix below). Denote by $G^\sharp$ the subgroup of G generated by its k -subtori. The group G is quasi-split (Lang) and therefore admits a Killing pair (T, B) . By definition, one has $T \subset G^\sharp$; in particular T normalizes $G^\sharp$. The big cell $R_u(B^-) T R_u(B)$ of G indicates that it suffices to verify that $R_u(B) \subset G^\sharp$. Let S be the maximal split k -torus of T . One considers the relative root system $\Phi(G, S)$ and a base Δ_k . Given $\alpha \in \Phi(G, S)$, one denotes by $U_{(\alpha)}$ the unipotent subgroup associated to α (cf. A. Borel, *Linear Algebraic Groups*, second edition (1991), Springer, Prop. 21.9). Since the k -group $R_u(B)$ is generated by the unipotent k -groups $U_{(\alpha)}$ ($\alpha \in \Delta_k$), one is reduced to verifying that $U_{(\alpha)} \subset G^\sharp$. A glance at the classification shows that there exists a semisimple group G_α of quasi-split type A_1 , 2A_1 , 3A_1 or 2A_2 such that $U_{(\alpha)} \subset G_\alpha \subset G$. It is therefore permissible to suppose that $G = \text{PGL}_2$ or $G = \text{SU}_3(K)$, where K denotes the unique quadratic field extension of k . The group $\text{PGL}_2^\sharp$ containing the standard split torus T , the possibilities up to conjugacy under $G(k)$ are the following: $\text{PGL}_2^\sharp = T$, $\text{PGL}_2^\sharp = B$, or $\text{PGL}_2^\sharp = \text{PGL}_2$. The case $\text{PGL}_2^\sharp = T$ is excluded since $\text{PGL}_2^\sharp$ contains the k -torus $R_{K/k}(\mathbf{G}_m)/\mathbf{G}_m$. The preceding discussion indicates that if $\text{PGL}_2^\sharp = B$, then $\text{PGL}_2^\sharp = \text{PGL}_2$. One therefore has $\text{PGL}_2^\sharp = \text{PGL}_2$. If $G = \text{SU}_3(K)$, $\Delta_k = \{\alpha\}$ and the possibilities for $G^\sharp$ (up to conjugacy) are the following: $G^\sharp = T$, $G^\sharp = B$, $G^\sharp = U_{(2\alpha)} \rtimes T$, $G^\sharp = \langle U_{(2\alpha)}, U_{(-2\alpha)}, T \rangle = \text{SL}_2 \cdot T$, or $G^\sharp = G$. The case $G^\sharp = B$ is excluded as for PGL_2 . The case $\text{SL}_2 \cdot T$ is excluded because $G^\sharp$ contains the k -torus $R_{K/k}^\sharp(R_{K_3/K}^1(\mathbf{G}_m))$, K_3/K denoting the extension of degree 3 of K , and the isogeny class of this torus is irreducible. Moreover, this torus also excludes the cases $G^\sharp = T$ and $G^\sharp = U_{(2\alpha)} \rtimes T$. One concludes that $G^\sharp = G$.

(1) This theorem is due to Chevalley. The writer wishes to express his gratitude to American Express which, by mislaying a trunk of Chevalley’s manuscripts, obliged him to reconstruct the proof.

(One says that G is geometrically simple if $G \otimes \bar{k}$ contains no normal group subscheme smooth over $\bar{k}$ apart from G and $\{e\}$; an equivalent condition: the associated root system R is irreducible.)

One may first suppose G indecomposable over k . The group $G \otimes \bar{k}$ is then a product of geometrically simple components S which are permuted transitively by the Galois group $\mathcal{G}$. If $\mathcal{H}$ is the subgroup of $\mathcal{G}$ fixing one of these components S , this component is defined over the field K corresponding to $\mathcal{H}$ (i.e. comes by extension of scalars from a subscheme of $G \otimes K$), and a standard argument shows that $G = \prod_{K/k}(S)$ (i.e. $R_{K/k}(S)$, for readers accustomed to Weil's notation). Likewise, the Lie algebra $\mathfrak{g}$ of G is identified with $\prod_{K/k} \mathfrak{s}$, where $\mathfrak{s}$ is the Lie algebra of S . If the theorem is true for S , there exists $x \in \mathfrak{s}(K) = \mathfrak{g}(k)$ which is regular in $\mathfrak{s}$; one then easily verifies that it is regular in $\mathfrak{g}$. Q.E.D.

From now on, one supposes G geometrically simple and chooses a maximal torus T of G (one knows that this is possible). Denote by X the character group of $T \otimes \bar{k}$, by R its root system, by W its Weyl group, and by E the group of automorphisms of X preserving R . The group W is a normal subgroup of E .

If T' is another torus of G , denote by X', R', W', E' the corresponding character group, root system, If one chooses $y \in G(\bar{k})$ such that $y \cdot (T \otimes \bar{k}) \cdot y^{-1} = T' \otimes \bar{k}$, one may identify X', R', W', E' with X, R, W, E by means of $\text{int}(y)$. Changing y changes this identification by an automorphism of X corresponding to an element of W .

The canonical generator $x \mapsto x^q$ of $\mathcal{G}$ acts on X' leaving R' stable; it therefore defines an element $f_{T'}$ of E' . One will in particular set $f = f_T$. When one identifies E' with E as just described, the element $f_{T'}$ becomes an element f' of E , defined up to replacement by $wf'w^{-1}$, with $w \in W$. One will compare this element to f :

Lemma 2. *One has $f' \equiv f \pmod{W}$; conversely, every $f' \in E$ satisfying this condition can be obtained from a maximal torus T' of G .*

Set $\bar{T} = T \otimes \bar{k}$, $\bar{T}' = T' \otimes \bar{k}$, and let $y \in G(\bar{k})$ be such that $y\bar{T}y^{-1} = \bar{T}'$; since $y^q\bar{T}y^{-q} = \bar{T}'$, one concludes that $n = y^{-1}y^q$ belongs to the normalizer $N(\bar{T})$ of $\bar{T}$. The effect of f' on the points of $\bar{T}(\bar{k})$ is then as follows:

$$f'(t) = y^{-1}(yty^{-1})^q y = y^{-1}y^q t^q y^{-q} y = nt^q n^{-1}.$$

If $w \in W$ is the element defined by n , this shows that f' and $f \circ w$ have the same effect on the points of $\bar{T}$, hence also on its characters, and one has $f' = f \circ w$, whence $f' \equiv f \pmod{W}$. Conversely, if $w \in W$ is given, one represents it by an element $n \in N(\bar{T})(\bar{k})$; by a classical theorem of Lang, one may write n in the form $n = y^{-1}y^q$, with $y \in G(\bar{k})$; the torus $\bar{T}' = y\bar{T}y^{-1}$ is then defined over k , and the preceding calculation shows that the corresponding f' is equal to $f \circ w$.

Lemma 3. *Let X, R, W, E be as above (R being irreducible), and let $\varphi \in E/W$. There then exists an element $f \in E$ representing φ , and a family $\theta_1, \dots, \theta_n$ of roots having the following two properties:*

- (1) $(\theta_1, \dots, \theta_n)$ is a basis of X .
- (2) R is the union of the orbits of the θ_i under the powers of f (i.e. every $a \in R$ is written $a = f^m \theta_i$, with suitable m and i).

The proof will be given a little later.

Here is now a linear algebra lemma:

Lemma 4. *Let V be a vector space over k , and $(\theta_1, \dots, \theta_n)$ a basis of the dual of $V \otimes \bar{k}$. There exists $x \in V$ such that $\theta_i(x) \neq 0$ for every i .*

Let V^* be the dual of V , and $\bar{W}_i$ the subspace of $V^* \otimes \bar{k}$ generated by the conjugates of θ_i ; this subspace is defined over k , i.e. of the form $W_i \otimes \bar{k}$. The obvious map $W_1 \otimes \dots \otimes W_n \longrightarrow \bigwedge^n V^*$ is nonzero (otherwise its extension to $\bar{k}$ would also be zero, which is absurd since

$\theta_i \in \bar{W}_i$, and $\theta_1 \wedge \cdots \wedge \theta_n \neq 0$). There therefore exists a basis of V^* consisting of elements $u_i \in W_i$. Let $x \in V$ be such that $u_i(x) \neq 0$ for every i (for example $u_i(x) = 1$). The element x answers the question, for if one had $\theta_i(x) = 0$, the conjugates of θ_i would also vanish at x , and the same would hold for u_i , which is not the case.

End of the proof of the theorem. Combining lemmas 2 and 3, one may choose a torus T whose element f satisfies the properties of lemma 3. If Y is the dual of X , the Lie algebra $\mathfrak{t}(\bar{k})$ of $\bar{T}^{(9)}$ is canonically identified with $Y \otimes \bar{k}$, and this operation is compatible with the action of the Galois group (the latter acting on $Y \otimes \bar{k}$ by its action on Y and on $\bar{k}$). Let $V = \mathfrak{t}(k)$ be the Lie algebra of T over k . An element $x \in V$ is regular if and only if it is annihilated by no root $\alpha \in R$, or rather by none of the linear forms $\bar{\alpha} \in V^* \otimes \bar{k}$ defined canonically by the $\alpha \in R$. By lemma 4, one may find such an x which is annihilated by none of the roots θ_i ; but every root is conjugate to a θ_i (this is what condition (2) of lemma 3 expresses); it follows that x is annihilated by no root, and it is indeed a regular element.

Proof of lemma 3. It relies on the properties of *Coxeter transformations*. Let us briefly recall what this consists of:

Let $a_1, \dots, a_n$ be a simple system of roots of R , and for every i let r_i be the reflection corresponding to a_i . Set:

$$c = r_1 \cdots r_n.$$

One has $c \in W$; of course, the element c depends on the choice of the simple system $(a_1, \dots, a_n)$ as well as on the order of the a_i ; however, one proves that its conjugacy class depends on none of these choices. One calls c the *Coxeter element* of the system under consideration. One proves (we shall admit it) that c does not have 1 as eigenvalue.

Lemma 5. *Set $\theta_i = r_n \cdots r_{i+1}(a_i)$. Then:*

- (a) *The θ_i form a basis of the group X generated by the roots.*
- (b) *One has $\theta_i > 0$ and $c(\theta_i) < 0$ for every i .⁽²⁾*
- (c) *Every root $a \in R$ such that $a > 0$ and $c(a) < 0$ is equal to one of the θ_i .*
- (d) *R is the union of the orbits of the θ_i under the powers of c .*

It is clear that θ_i is of the form $a_i + \sum_{j>i} m_{ij}a_j$, with $m_{ij} \in \mathbf{Z}$, which proves (a). Assertions (b) and (c) are consequences of the following observation: the reflection r_i preserves the sign of every root distinct from $\pm a_i$, and changes the sign of $\pm a_i$.

Finally, for (d), one observes that an orbit of c cannot consist entirely of positive (resp. negative) roots, for by taking the sum of these roots one would find a nonzero element of X invariant under c , and one has been willing to admit that c does not have 1 as eigenvalue. There is therefore necessarily in every orbit an element $a > 0$ such that $c(a) < 0$, and one applies (c).

Remark. The preceding proof has been sketched only to facilitate the reader's task; one could have confined oneself to referring to the canonical texts on Coxeter (cf., for example, Koszul, Séminaire Bourbaki, 1959/1960, exposé 191). These texts contain other results: the orbits of c all have the same number h of elements, and each contains only one θ_i . In particular, $h = \text{Card}(R)/n$.

Let us now return to the proof of lemma 3. Distinguish three cases:

- (1) *The given element $\varphi \in E/W$ is equal to the identity element.*
One must then take $f \in W$; one chooses $f = c$; this works by lemma 5.
- (2) *The system R has type A_n , with even $n \geq 2$, and φ is the unique nontrivial element of E/W .*

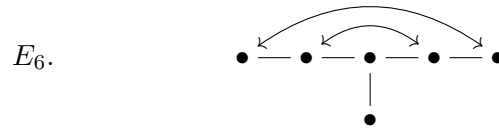
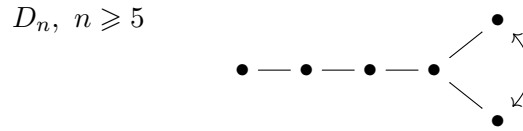
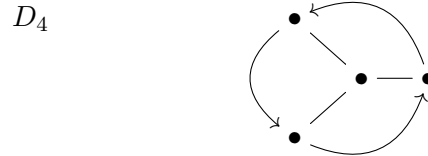
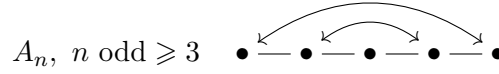
One knows that $-1 \notin W$; one then takes $f = -c$. A simple calculation shows that c has order $h = n + 1$; its order is therefore odd. If $a \in R$ is any root, one has $a = c^m \theta_i$ for a pair

⁽⁹⁾**N.D.E.** T has been changed to $\bar{T}$.

⁽²⁾For the order relation defined by $(a_1, \dots, a_n)$.

(m, i) ; cf. lemma 5; adding h to m if necessary, one may suppose m even, and one sees that one then has $a = (-c)^m \theta_i = f^m \theta_i$. The orbits of the θ_i therefore indeed fill R .

(3) The element $\varphi \in E/W$ is nontrivial, and R has one of the following types:



(A glance at the classification shows that these are indeed all the cases (with A_n , even n) where E/W is nontrivial.)

Let S be a simple system of roots, which we do not number for the moment. One knows that E is the semidirect product of W and the group Ψ of permutations of S which leave the Cartan matrix invariant (or the Dynkin diagram; it is the same thing). The group E/W is thus identified with Ψ , and in particular φ corresponds to an element $\psi \in \Psi$. One observes by inspecting the Dynkin diagrams (cf. the figures above) that every orbit of ψ in S consists of pairwise orthogonal roots (i.e. not joined in the diagram). [Note that this would not be the case for A_n (even n), which obliged us to treat this case separately.] If σ is such an orbit, the reflections $r_a, a \in \sigma$, commute with one another; their product will be denoted by ρ_σ . It is clear that ρ_σ commutes with ψ .

This being so, let us choose a total order on S such that every orbit is a segment for this order relation; this amounts to numbering the elements of S : $S = \{a_1, \dots, a_n\}$. Let $\theta_1, \dots, \theta_n$ be the roots defined above, and $c = r_1 \cdots r_n$ the corresponding Coxeter element. The element c is a product of the ρ_σ , the σ being arranged in a certain order; it follows that it commutes with ψ . One then sets $f = c\psi$. One observes in addition that ψ permutes the θ_i among themselves. Indeed, one has $\psi(\theta_i) > 0$ since $\theta_i > 0$, and $c(\psi(\theta_i)) = \psi(c(\theta_{i'})) < 0$, hence (lemma 5, (c)), $\psi(\theta_i)$ is equal to a θ_j . It is now immediate that $f = c\psi$ answers the question. Indeed, if $a \in R$, one has $a = c^m \theta_i$ for a pair (m, i) , whence $a = f^m(\psi^{-m} \theta_i) = f^m \theta_j$ for a certain j . Q.E.D.

Remark. One can prove that, except in case (2), every orbit of f has exactly $h = \text{Card}(R)/n$ elements and contains one and only one θ_i . In case (2), some of the θ_i are unnecessary.